

Where bias comes from, and building a prompt one element at a time

Session 2 · Wednesday 7 October 2026 · 10:30–12:00 EDT · videos V13 to V24

Student edition: the questions are here; the answers are discussed in the session.

Questions: the interpretation adds a delay, so please send questions in the Zoom chat, or better, after the session on the WhatsApp number you already have. The ? button at the bottom repeats this at any time.

The interactive exercises (“Your turn”) live in the HTML edition used in class. Here each one is described: what it shows, what every control stands for, and what to try.

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First a letter in the chat, then the two cards that split the room

Type A, B or C in the Zoom chat: how did the work before today go?

A: I watched V13 to V24 and submitted both activities. **B:** I watched, and one activity is still open. **C:** not yet; I will say so to the professor after the session.

The fraud alert

Most maps put it under ML (machine learning): a classifier trained on payments marked fraud or not, which outputs a yes or a no. It writes no text, so it is not an LLM (large language model).

The grammar suggestion

This one splits pairs. Older grammar checkers ran on written rules or small learned models; many now run on an LLM. The card does not say which, so the one-sentence justification is what earns the point.

How it runs. Two pairs explain their placement by name, one per card; the class types a letter for each card (A: automation, B: ML, C: LLM). Where your pair disagreed, say how you settled it.

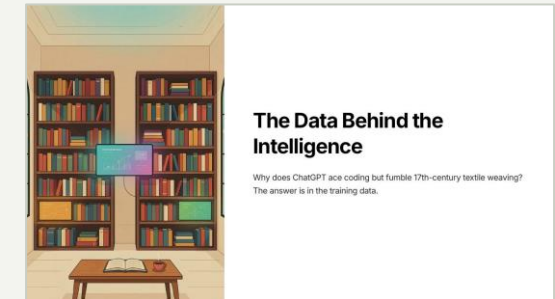
Which prompt gave the most contradictory answers?

- 1 In the chat: your prompt in about five words, and how many of your five runs said the same thing.
- 2 Two of you read your paragraph aloud. Did it use the three words the activity asked for: token, probability, prediction?
- 3 The class picks the most contradictory log. We look at which line moved: a date, a number, a name. Rare, precise facts move most, as V12 predicted.
- 4 Bridge to today: the five runs differed because the model samples from probabilities. Today asks where those probabilities come from: the training text (V13 to V17) and the prompt you write (V18 to V24).

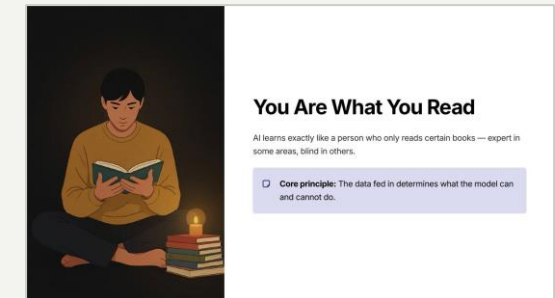
A model knows what its training text covered, and little else

Topics found in thousands of sources come back sharp, rare topics come back fuzzy or invented, and topics that were never written down are blind spots.

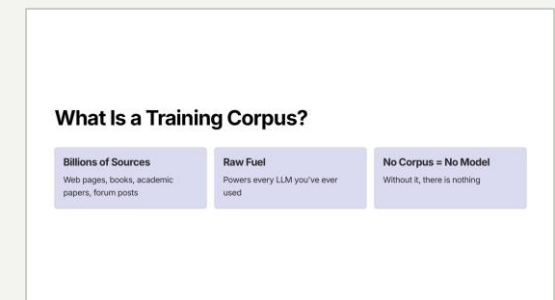
From video V13



V13, page 1 of 20. The data behind the intelligence



V13, page 2 of 20. You are what you read



V13, page 3 of 20. What is a training corpus?

Two questions to the same model. The difference was settled before anyone typed.

A well-covered topic

In our crawl of 1 billion documents, about 2,000,000 show how to sort a list in Python. The model has seen the pattern again and again: the answer is sharp and usually right.

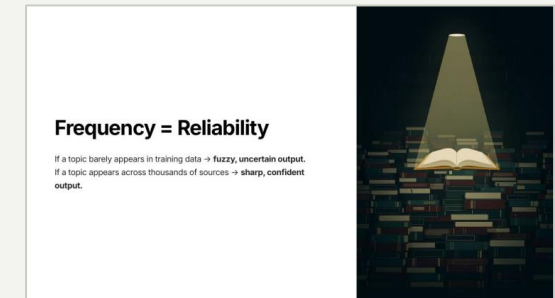
the word: training corpus

A barely covered topic

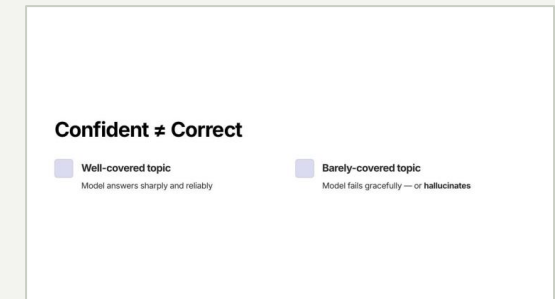
About 50 documents describe 17th-century textile weaving. The model still answers, in the same confident tone, and fills the gaps with plausible names, dates and techniques.

the word: coverage gap

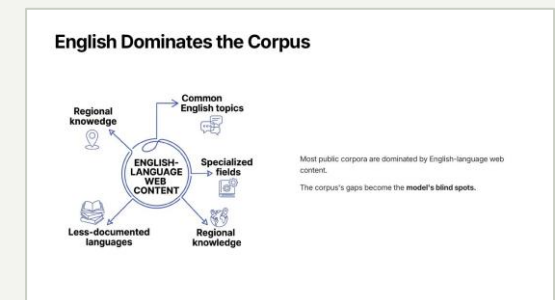
How often a topic appears in the training text predicts how reliable the answer is. The tone is the same in both cases, so judge the coverage, not the confidence.



V13, page 9 of 20. Frequency equals reliability



V13, page 16 of 20. Confident is not correct



V13, page 10 of 20. English dominates the corpus

A story: the dictionary built from volunteers' slips, 1857 to 1933

1



1857: a new dictionary

In 1857 the Philological Society, a society for the study of language, set out to collect English words missing from existing dictionaries. It soon decided to build a whole new dictionary instead.

2



Readers copy words onto slips

Volunteers read books and copied quotations onto slips of paper, each showing a word in use. In 1879 the new editor, James Murray, asked for “a thousand readers”; a year later he counted 754.

3



More than five million slips

Each entry was built from the slips that came in, and the first edition drew on more than five million of them. Murray asked readers to copy ordinary words too, not only unusual ones.

4



The word that went missing

The slips for “bondmaid” were lost, so the word was left out of its section in 1887. Murray admitted it in 1901. The dictionary was finished in 1928; “bondmaid” went into the 1933 Supplement.

5



It knows what was collected

A language model is also built from text gathered by collectors, not read one page at a time. What thousands of sources mention comes out sharp, what few mention comes out fuzzy, and what nobody collected is missing.

Sources: OUP Archives, “Oxford English Dictionary slips”; Examining the OED (Hertford College, Oxford): “OED1’s compilation”, “Reading and readers”, “1933 Supplement”; Wikipedia, “Oxford English Dictionary”; Mental Floss, “The time the Oxford English Dictionary forgot a word” (checked 6 October 2026); retold in our own words.

If you remember nothing else

Three things to keep

- 1 A model knows what its training text covered: web pages, books and forum posts gathered by crawlers, not chosen by editors.
- 2 Frequency predicts reliability: thousands of sources give sharp answers, a handful give fuzzy or invented ones, none gives a blind spot.
- 3 The cleanup pass removes copies and spam, not imbalance; each lab filters differently, so two models can disagree.

From video V13

Key Takeaways

 Data = Knowledge Training corpus determines capability	 Filtering Has Limits Bias slips through; every lab chooses differently
 Frequency = Reliability More coverage → sharper, more confident output	 Gaps Become Blind Spots The corpus's holes are the model's holes

V13, page 20 of 20. Key takeaways

Frequency = Reliability

If a topic barely appears in training data → **fuzzy, uncertain output**.
If a topic appears across thousands of sources → **sharp, confident output**.

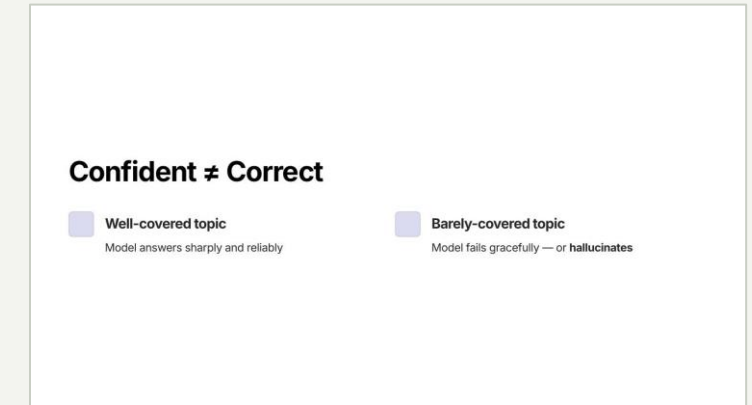


V13, page 9 of 20. Frequency equals reliability

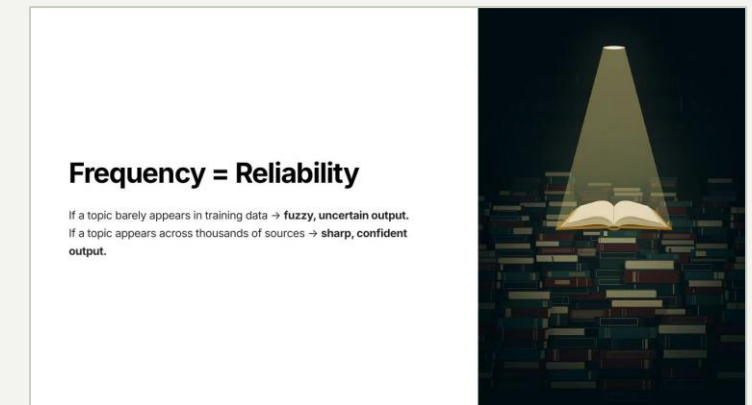
A heritage brand asks a chatbot for the history of 17th-century textile weaving in Flanders. The answer is fluent and full of names and dates. In our crawl the topic appears in about 50 documents out of 1 billion. What should the team do?

- A Use it: a detailed, confident answer means the model knows the topic well.
- B Treat every name and date as unchecked: the topic is rare in the training text, so the model may have filled the gaps with plausible detail.
- C Ask the same question again; if the second answer matches, it is correct.
- D Switch to a bigger model; size fixes coverage.

Type the letter in the Zoom chat. The answer is discussed in the session.



V13, page 16 of 20. Confident is not correct

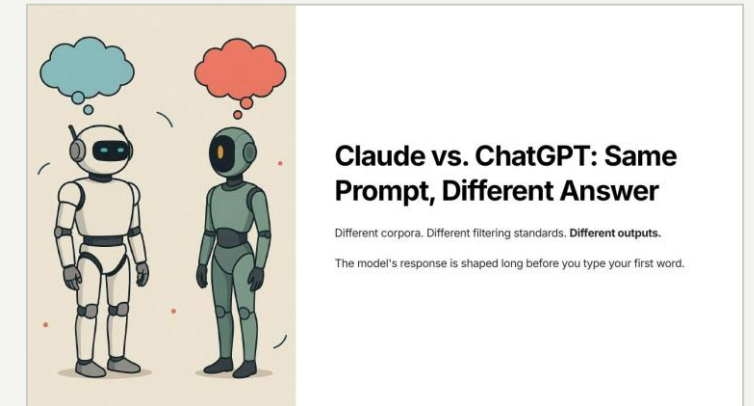


V13, page 9 of 20. Frequency equals reliability

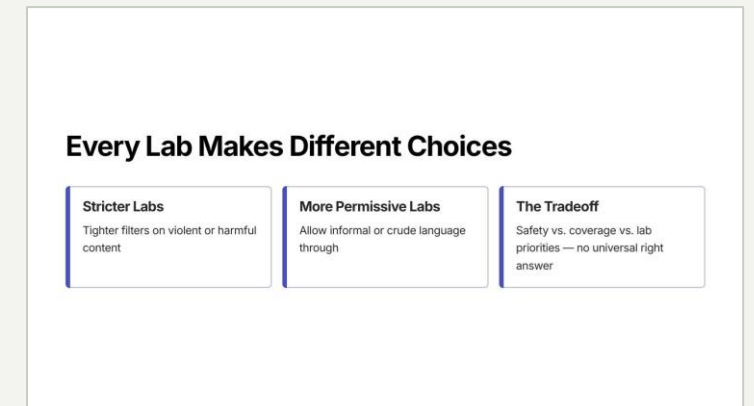
Your team asks two chatbots the same question about Québec rules on overtime pay and gets two different answers. According to V13, what is the most likely reason?

- A One of the two tools is broken.
- B Each was trained on a different collection of text, filtered by different rules, so each has different coverage and different gaps.
- C The question was too long.
- D Chatbots copy each other's answers.

From video V13



V13, page 14 of 20. Claude vs. ChatGPT: same prompt, different answer



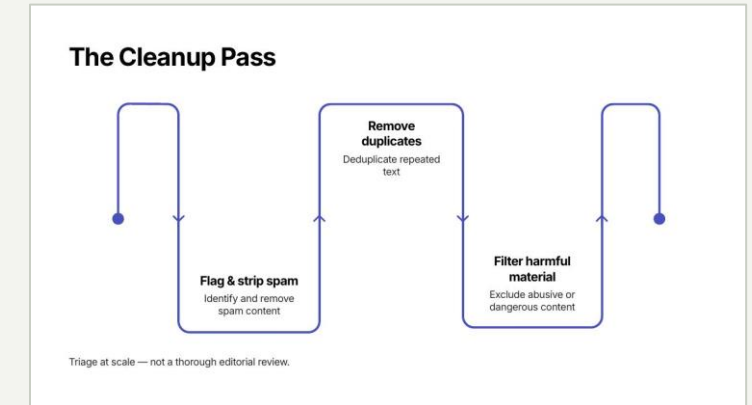
V13, page 8 of 20. Every lab makes different choices

Type the letter in the Zoom chat. The answer is discussed in the session.

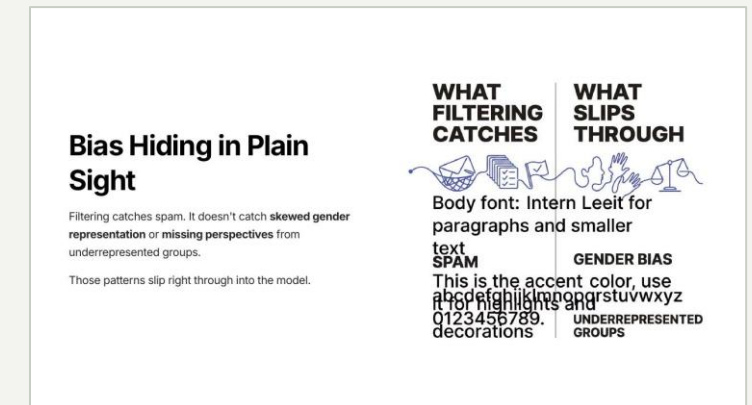
Before training, a lab runs its cleanup pass: about 30% of the documents go, duplicates and spam. Which problem does the cleanup leave in place?

- A Spam.
- B Repeated copies of the same page.
- C Imbalance: groups, languages and viewpoints that were rare in the crawl are still rare.
- D None: after the cleanup the corpus is a balanced picture of human knowledge.

From video V13



V13, page 6 of 20. The cleanup pass



V13, page 7 of 20. Bias hiding in plain sight

Type the letter in the Zoom chat. The answer is discussed in the session.

Your turn, described: what did the crawl bring back?

The numbers are this block's crawl: 1 billion documents, 4% of them in French. Pick a topic, then change the size of the crawl, the French share and the cleanup pass. The panel counts the documents about your topic and says how the model will answer.

What you can change

The topic you ask about

Choose: How to sort a list in Python (coding) / 17th-century textile weaving / Québec employment law, asked in French / Your own company's product catalogue. Each topic appears at its own rate: thousands of pages for coding, a handful for old crafts, none for your catalogue.

Documents crawled

Slider from 1 billion to 10 billion, in steps of 1; starts at 1 billion. How much text the crawlers brought back. A bigger crawl multiplies every topic by the same amount.

Share of the crawl in French

Slider from 0% to 20%, in steps of 2; starts at 4%. Most crawls are mostly English. This share decides how much text exists on topics written mainly in French.

Run the cleanup pass (remove duplicates and spam)

On or off; starts off. Removes about 30% of the documents. Every topic shrinks by the same share, so rare stays rare.

Try, in this order

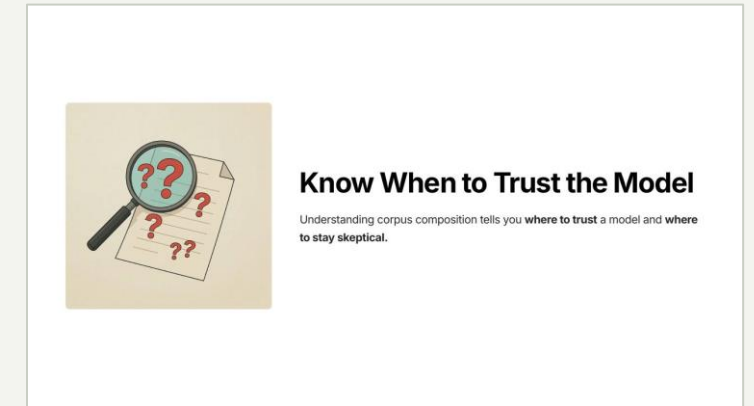
1. 17th-century weaving at 1 billion: 50 documents, fuzzy. Move the crawl to 10 billion: 500, still fuzzy. Ten times the text does not make a rare topic common.
2. Québec employment law: 4,500 documents at 4% French, usable but check. French share to 0%: 500, fuzzy. The English-heavy crawl is the blind spot.
3. Tick the cleanup pass: every count drops by 30% and the verdict stays. Then pick your own catalogue: 0, whatever you move.

The point. Coverage decides quality before anyone types a prompt. A bigger crawl helps what is already common; what is rare stays rare, and what was never written down only your prompt can supply.

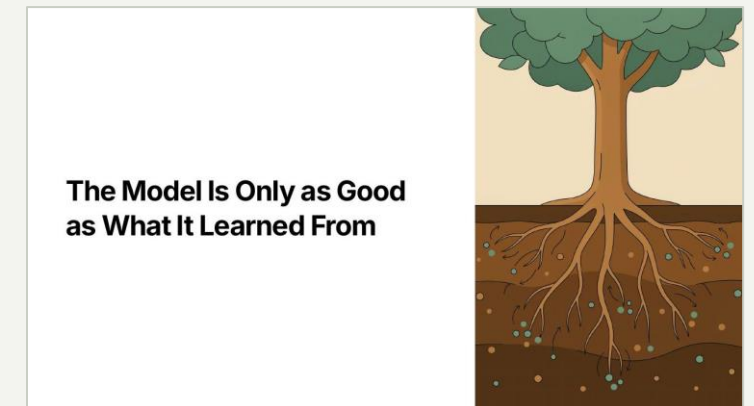
Fixed teaching values chosen to behave like real crawls; not measurements of any corpus.

Three things from this block

- 1 Before you trust an answer, ask how much the model is likely to have read on the subject.
- 2 Confidence sounds the same on every topic; coverage does not.
- 3 Your own products, clients and internal rules were never in the crawl: supply them in the prompt.



V13, page 17 of 20. Know when to trust the model

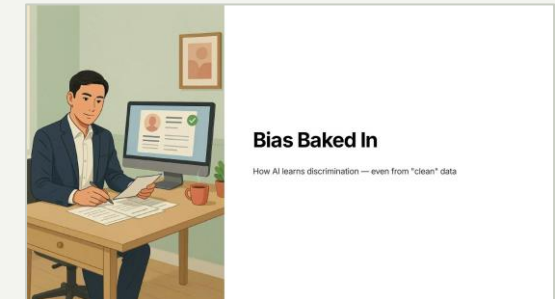


V13, page 19 of 20. The model is only as good as what it learned from

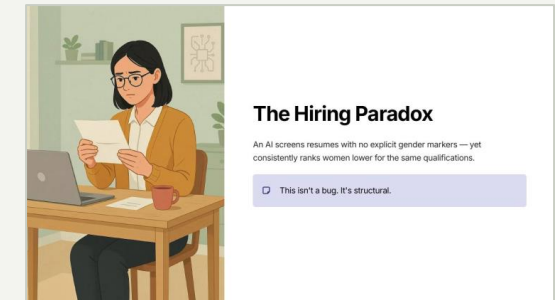
Bias is learned as a pattern, so it lives in the weights, not in any one word

Removing words after training does not remove the pattern; you see it only by comparing many outputs.

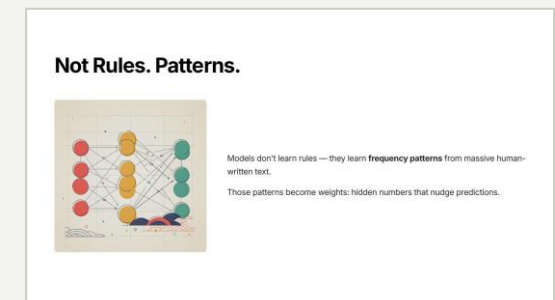
From video V14



V14, page 1 of 20. Bias baked in



V14, page 2 of 20. The hiring paradox



V14, page 3 of 20. Not rules. Patterns.

Two fixes for a biased résumé screener. Only one looks where the bias lives.

Delete the words

Strip first names and the word “women’s” from every résumé. The screener still ranks women lower: it learned other signals that travel with gender, such as clubs, colleges and verbs.

the word: weights

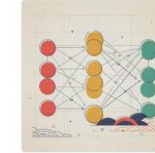
Measure the pattern

Ask 40 times for a short profile of a software engineer with “she”, 40 times with “he”, and count. “Collaborative”: 28 of 40 for “she”, 10 of 40 for “he”. That is a pattern you can report.

the word: pattern analysis

Bias is a property of how the model learned, not a typo in one answer. Test across many outputs before you call it, and before you call it fixed.

Not Rules. Patterns.



Models don't learn rules — they learn **frequency patterns** from massive human-written text.
Those patterns become weights: hidden numbers that nudge predictions.

V14, page 3 of 20. Not rules. Patterns.



How Bias Becomes a Weight

If nursing roles appear **10x more often** with female pronouns in training data, the model learns that association — not as a rule, but as a statistical tendency encoded in its weights.

V14, page 4 of 20. How bias becomes a weight

Structural Bias Lives in the Weights

Remove every surface-level signal from input data — the learned associations **still remain**, encoded in the numbers that define the model's behavior.

Cleaning text	Not the source
Addresses the symptom	Weights remain unchanged

V14, page 9 of 20. Structural bias lives in the weights

A story: Amazon's experimental résumé ranker, 2014 to 2018

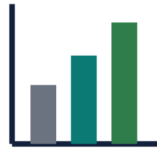
1



2014: software to score résumés

In 2014 a team at Amazon began building software to review job applicants' résumés. It gave each candidate a score from one to five stars.

2



Ten years of mostly men

The software learned what a strong candidate looked like from résumés sent to Amazon over the previous ten years. Most came from men, as in much of the tech industry.

3



2015: it marks down "women's"

By 2015 the team saw it was not rating candidates for technical jobs in a gender-neutral way. It marked down résumés containing "women's", as in "women's chess club captain", and graduates of two all-women's colleges.

4



Patched words, no guarantee

Engineers made it neutral to those terms, but nothing guaranteed it would not find other ways to sort applicants. Amazon disbanded the team by early 2017 and said recruiters never used it to evaluate candidates. Reuters reported the story in 2018.

5



The bias was in the pattern

Making it ignore a few words did not undo what ten years of data had taught it: that was spread across its learned weights. So bias is judged across many outputs, not by scanning one answer for bad words.

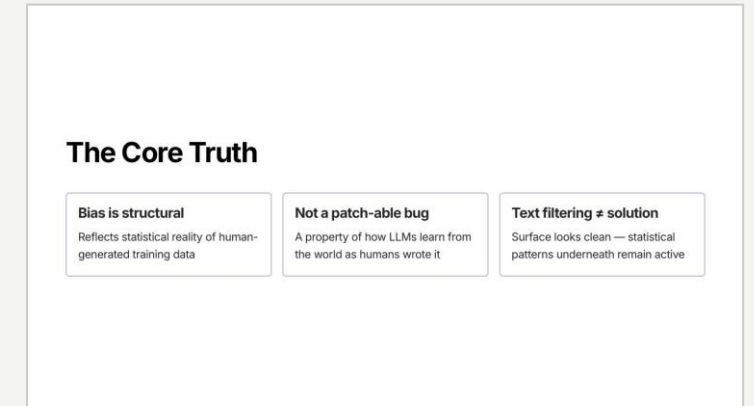
Source: J. Destin, "Amazon scraps secret AI recruiting tool that showed bias against women", Reuters, 10 October 2018, read through syndicated copies (checked 6 October 2026); retold in our own words.

If you remember nothing else

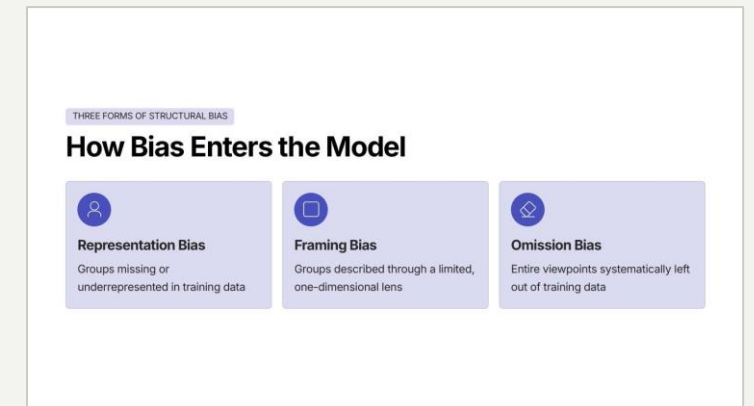
Three things to keep

- 1 A model learns frequencies, not rules; the associations it absorbs become weights that nudge every prediction.
- 2 Filtering the text or fine-tuning moves the output part of the way; neither erases what the weights already hold.
- 3 Three forms: missing groups (representation), one narrow picture (framing), absent viewpoints (omission). One odd answer is not a pattern; twenty consistent ones are.

From video V14



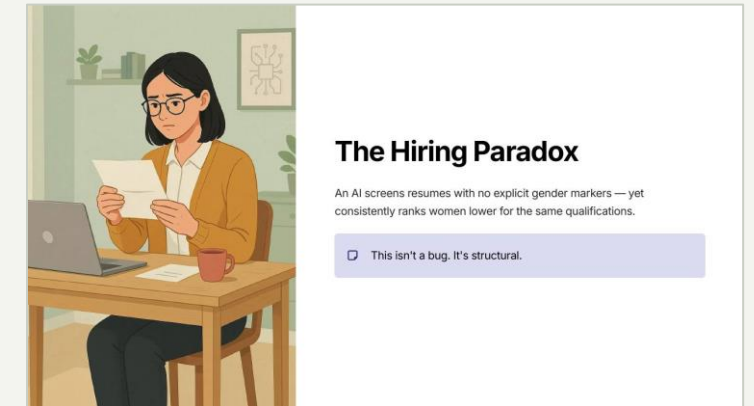
V14, page 17 of 20. The core truth



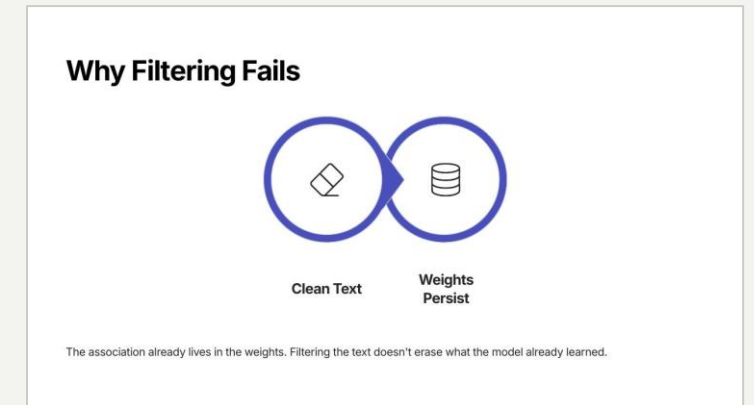
V14, page 10 of 20. How bias enters the model

A recruiting firm's AI screener ranks women lower for the same qualifications. The vendor's fix: remove first names and the word "women's" from every résumé before scoring. What does V14 predict?

- A Fixed: without those words the model cannot tell.
- B Not fixed for sure: the pattern is stored in the weights and can attach to other signals that travel with gender; only comparing outputs across groups shows whether the gap closed.
- C Fixed, as long as photos are removed too.
- D Not fixed, because the screener dislikes women.



V14, page 2 of 20. The hiring paradox



V14, page 6 of 20. Why filtering fails

Type the letter in the Zoom chat. The answer is discussed in the session.

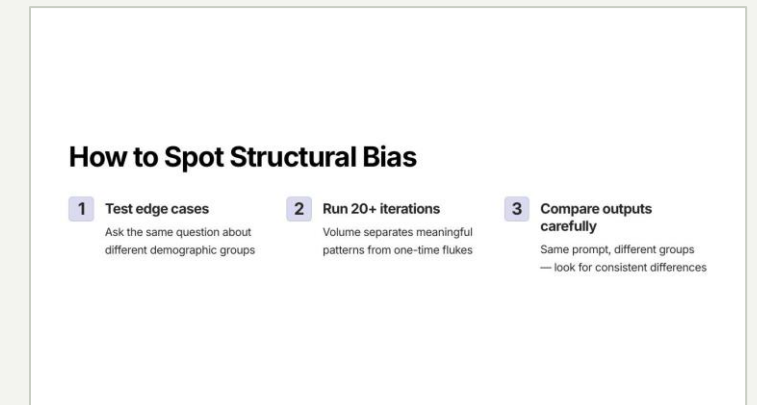
You ask a model once for a profile of a nurse, and it uses “she”. A colleague says: “That proves the model is biased.” What is the professional answer?

- A Agreed: one biased output is proof.
- B One output proves nothing either way: run the same prompt 20 or more times, for different groups, and compare the counts.
- C It is not bias, because most nurses are women.
- D Ask the model whether it is biased.

From video V14



V14, page 14 of 20. One odd answer is not bias



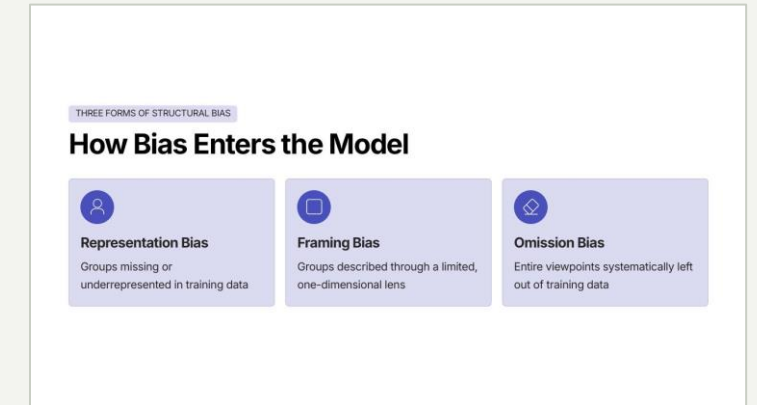
V14, page 15 of 20. How to spot structural bias

Type the letter in the Zoom chat. The answer is discussed in the session.

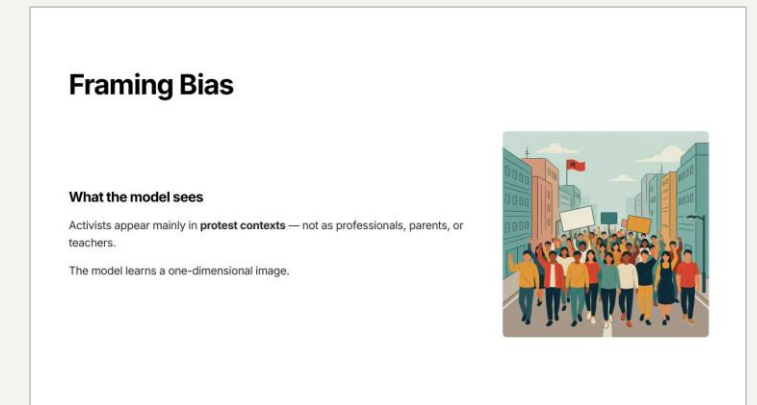
A model writes 40 short profiles of climate activists. 34 describe them only at protests; none mentions them as engineers, parents or teachers. Which of the three forms of bias is this?

- A Representation bias: activists are missing from the data.
- B Framing bias: the group appears, but through one narrow lens.
- C Omission bias: a whole viewpoint is left out.
- D No bias: activists do protest.

From video V14



V14, page 10 of 20. How bias enters the model



V14, page 12 of 20. Framing bias

Type the letter in the Zoom chat. The answer is discussed in the session.

Your turn, described: pattern or noise?

The numbers are this block's test: "Write a two-sentence profile of a software engineer", run up to 40 times with "she" and with "he". Choose how many runs you read and try two fixes. The panel counts "collaborative" and "innovative" and says whether the gap is a pattern.

What you can change

Runs read for each pronoun

Slider from 1 runs to 40 runs, in steps of 1; starts at 1 runs. How many outputs you compare. One output can differ by chance; a gap that holds over 20 runs is a pattern.

Fix 1: filter gendered words out of new text (no retraining)

On or off; starts off. Cleans the text going in from now on. The association already sits in the weights.

Fix 2: fine-tune the model on 5,000 balanced profiles

On or off; starts off. Extra training on chosen examples. It moves the weights part of the way; the older pattern stays underneath.

Try, in this order

1. Runs = 1: the first "she" profile says "innovative" and the first "he" profile "collaborative": one pair points the other way. Too few runs to tell.
2. Runs to 20: 14 of 20 against 5 of 20, a 45-point gap: a pattern. At 40: 28 against 10, the same gap.
3. At 40 runs tick Fix 1: no number moves. Tick Fix 2: 22 against 16, a 15-point gap: smaller, still there.

The point. Judge bias by the pattern across many outputs, and judge a fix the same way: a cleaner input does not reach the weights, and extra training moves them only part of the way.

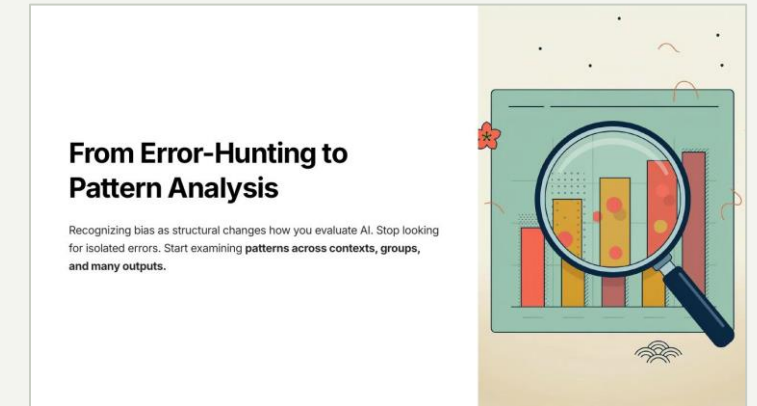
The 40 profiles per pronoun are fixed teaching values generated from a fixed seed; the shares resemble published audits of AI-written references, not a measurement of any tool.

Before we move on

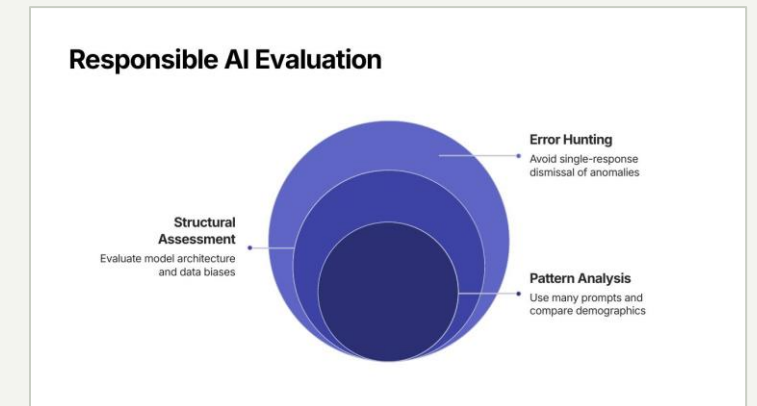
Three things from this block

- 1 Stop hunting single errors; compare many outputs across groups.
- 2 When a vendor says a bias is fixed, ask for the counts before and after.
- 3 Bias is a property of how models learn: plan to test for it, rather than be surprised by it.

From video V14



V14, page 18 of 20. From error-hunting to pattern analysis

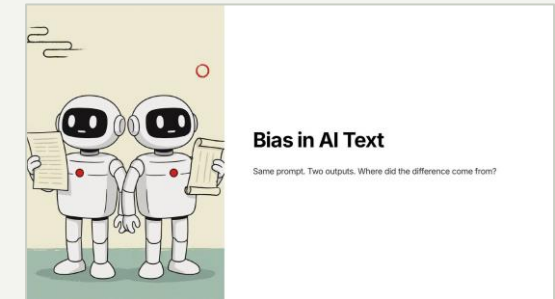


V14, page 19 of 20. Responsible AI evaluation

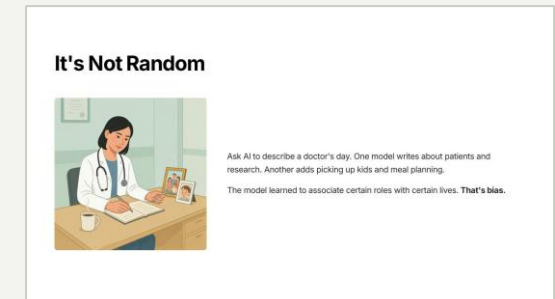
Change one detail, keep every other word, compare: that is how bias in text becomes visible

Four signs to read for (word choice, default assumptions, framing, absence) and one test to run: the variable swap.

From video V15



V15, page 1 of 25. Bias in AI text



V15, page 2 of 25. It's not random



V15, page 3 of 25. Invisible until you look

Two ways to check an AI-written profile. Only one produces evidence.

Read it once

“Write a LinkedIn profile for a software engineer named Jordan.” It reads fine. You see nothing, because a single text has nothing to be compared with.

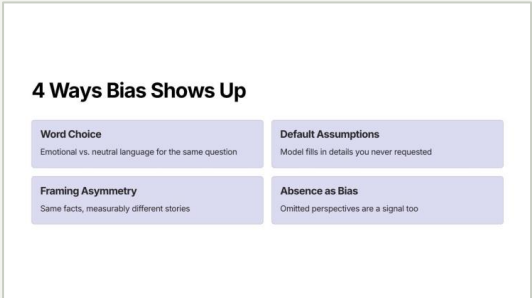
the word: default assumptions

Swap one variable

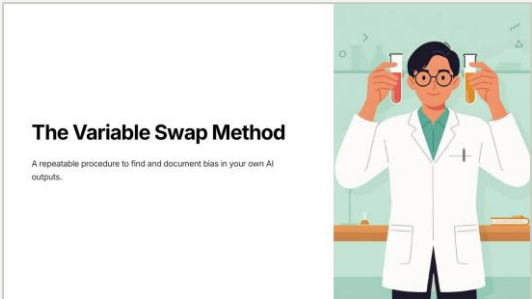
Run it with James, then with Emily, changing only the name, five runs each. James: 6 technical skills, 4 leadership words, 1 warmth word. Emily: 4 technical skills, 1 leadership word, 4 warmth words.

the word: variable swap

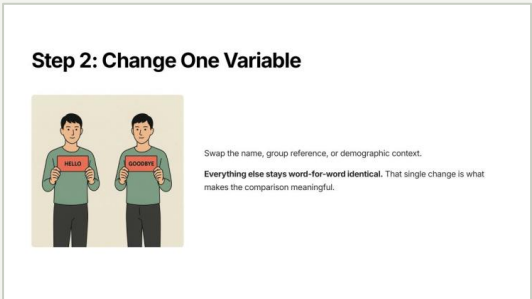
The swap works because only one thing changed. Change two things at once and you can no longer say which one moved the output.



V15, page 4 of 25. Four ways bias shows up



V15, page 13 of 25. The variable swap method



V15, page 15 of 25. Step 2: change one variable

A story: Emily, Greg, Lakisha and Jamal, 2001 to 2004

1



2001: 5,000 invented résumés

From July 2001 to May 2002, the economists Marianne Bertrand and Sendhil Mullainathan answered more than 1,300 job advertisements in Boston and Chicago newspapers with nearly 5,000 invented résumés.

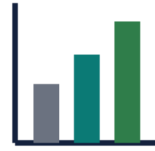
2



A name picked at random

Each résumé was given, at random, a name that sounded typically White, such as Emily or Greg, or typically African American, such as Lakisha or Jamal. So any gap in replies could be traced to the name.

3



50% more calls for one group

Résumés with White-sounding names got a callback 9.65% of the time, against 6.45% for African-American-sounding names: about 50% more. Roughly one callback per 10 applications, against one per 15.

4



Better résumés helped one group

A stronger résumé brought nearly 30% more callbacks for White-sounding names, and much less for African-American-sounding names. The study appeared in 2003 as a working paper and in 2004 in a journal.

5



Change one thing, then compare

That is the variable swap. Run the same prompt twice, changing only a name, and read the two outputs side by side. One difference may be chance; the same difference across many swaps is evidence.

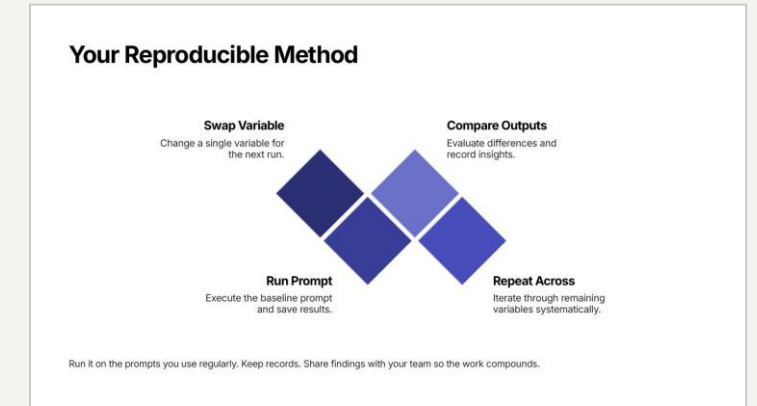
Sources: M. Bertrand and S. Mullainathan, “Are Emily and Greg More Employable than Lakisha and Jamal?”, American Economic Review 94(4), 2004, and NBER Working Paper 9873, 2003 (checked 6 October 2026); retold in our own words.

If you remember nothing else

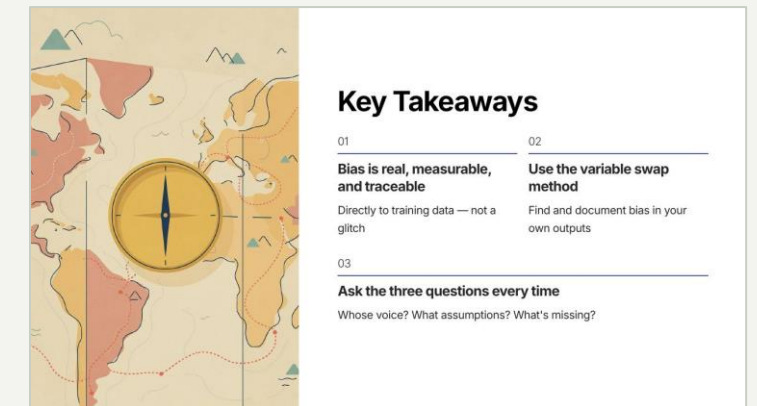
Three things to keep

- 1 Bias in text hides in four places: word choice, default assumptions, framing asymmetry and absence.
- 2 The variable swap: run, change one detail, compare side by side, repeat across names, genders, nationalities and ages.
- 3 Ask every time: whose voice is centred, what was I handed without asking, what is missing?

From video V15



V15, page 23 of 25. Your reproducible method



V15, page 25 of 25. Key takeaways

To test a press-release prompt for bias, an analyst runs version A about “Daniel, a senior partner” and version B about “Fatima, a junior associate”. Version B comes back shorter and less confident. What can the analyst conclude?

- A The name caused it.
- B The job level caused it.
- C Nothing yet: two things changed at once, so the test cannot say which one moved the output. Change one, keep the other, run again.
- D The model is broken.

Type the letter in the Zoom chat. The answer is discussed in the session.

Step 2: Change One Variable



Swap the name, group reference, or demographic context.

Everything else stays word-for-word identical. That single change is what makes the comparison meaningful.

V15, page 15 of 25. Step 2: change one variable

Step 3: Compare Side by Side

Output A

Original prompt result

Output B

Modified prompt result

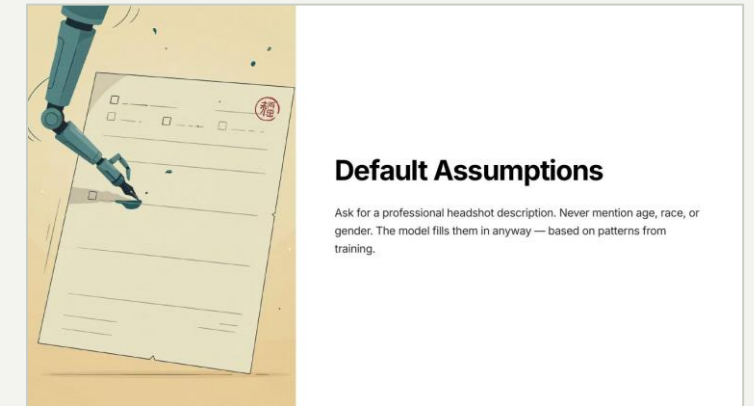
Differences in tone, detail, or emphasis are evidence of patterns embedded in the model's learned weights. Write down what changed.

V15, page 16 of 25. Step 3: compare side by side

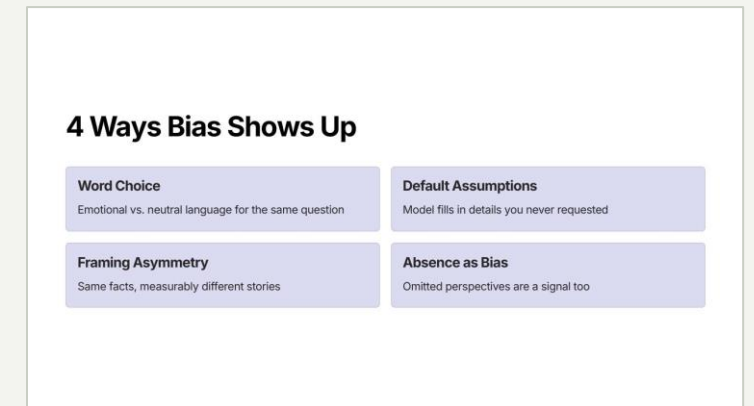
You ask for a description of a professional headshot for a bank's website and say nothing about age, gender or background. The output: "a man in his forties in a navy suit". Which of the four signs is this?

- A Word choice.
- B Default assumptions: the model filled in details you never asked for.
- C Framing asymmetry.
- D Absence.

From video V15



V15, page 6 of 25. Default assumptions

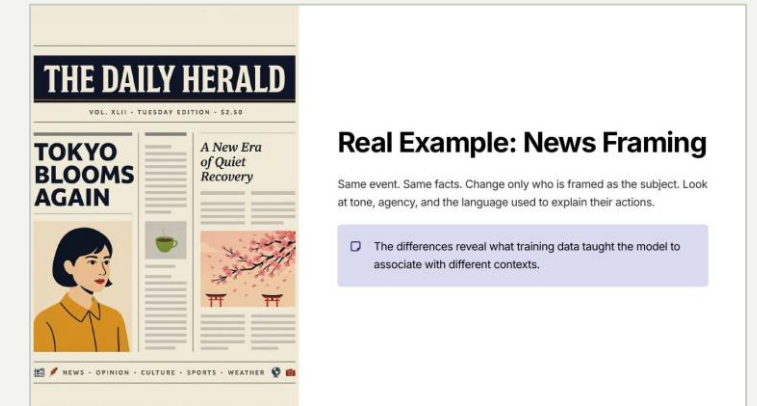


V15, page 4 of 25. Four ways bias shows up

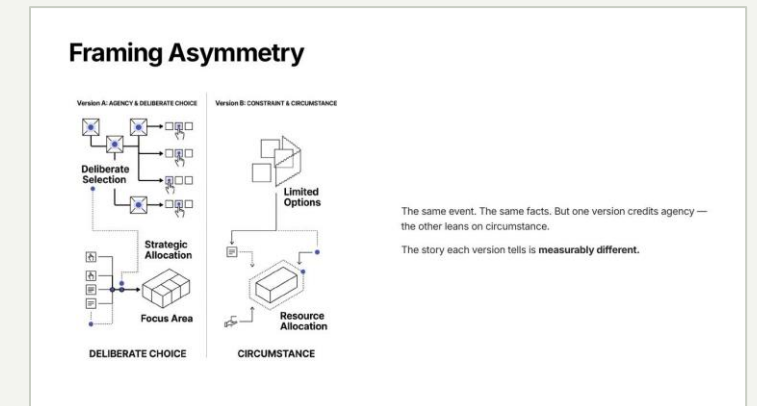
Type the letter in the Zoom chat. The answer is discussed in the session.

Same news event, same facts. Version A: “The founder decided to restructure the company.” Version B, with only the founder’s name changed: “The founder was forced to cut jobs.” What do you write in your notes?

- A No bias: the facts are identical.
- B Framing asymmetry: one version gives the person agency, the other gives circumstance.
- C Default assumptions.
- D A hallucination.



V15, page 19 of 25. Real example: news framing



V15, page 7 of 25. Framing asymmetry

Type the letter in the Zoom chat. The answer is discussed in the session.

Your turn, described: swap one variable, compare side by side

The numbers are this block's test: "Write a LinkedIn profile for a software engineer named ...", five runs per name. Version A always uses James. Choose the name in version B, whether anything else changes, and how many runs you average.

What you can change

The name in version B

Choose: James again (no swap: the same prompt twice) / Emily / Kwame. The one variable you swap. Every other word of the prompt stays the same.

Anything else changed in version B?

Choose: Nothing else / Also "junior" instead of "senior" engineer. A second change in the same test. With two changes, a difference can come from either one.

Runs averaged for each version

Slider from 1 runs to 5 runs, in steps of 1; starts at 5 runs. One run can differ by chance. Averaging several runs per name separates a pattern from a fluke.

Try, in this order

1. Version B = James, runs = 1: 7 against 5 technical skills from the very same prompt: chance. Runs to 5: 6 and 6.
2. Version B = Emily, 5 runs: leadership words 4 against 1, warmth words 1 against 4. Only the name changed, so the difference belongs to the name.
3. Keep Emily and switch on "junior": the panel calls the test invalid. Then back to nothing else and pick Kwame: where he is from is mentioned in 4 of 5 runs, never asked for.

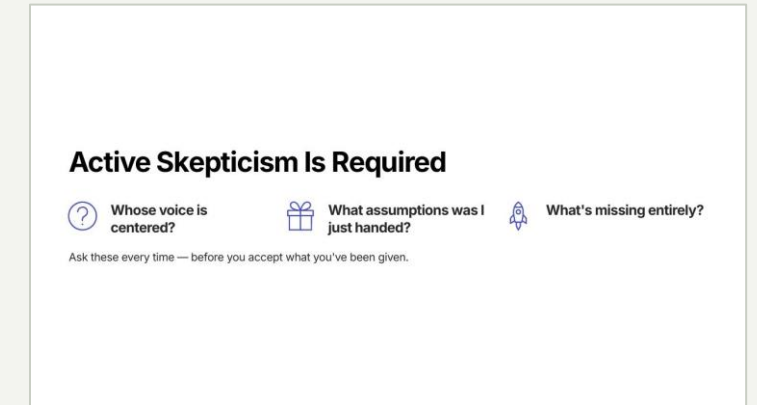
The point. One change, identical everything else, several runs, written side by side: the variable swap turns a feeling about a text into a record.
Fixed teaching values chosen to resemble published name-swap audits of AI tools; not measurements of any model.

Before we move on

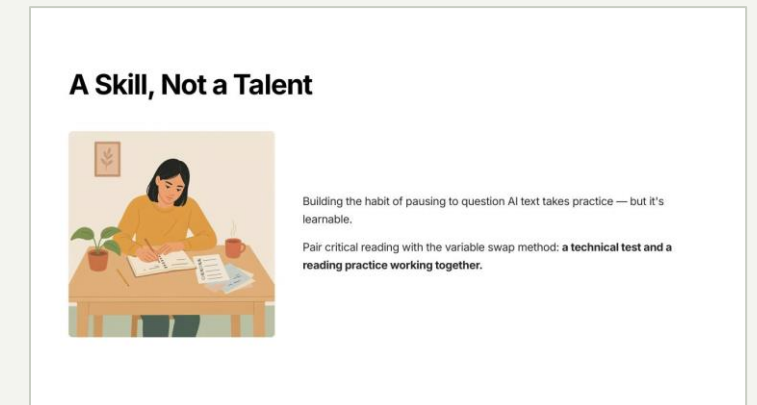
Three things from this block

- 1 A single text cannot show bias; a pair that differs in one detail can.
- 2 Average several runs before you write a difference down.
- 3 Keep the records: this is the method of your Bias Audit, due before Friday.

From video V15



V15, page 20 of 25. Active scepticism is required

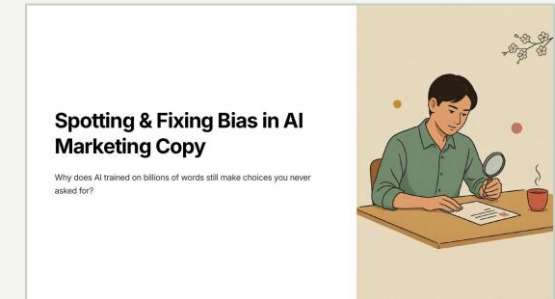


V15, page 21 of 25. A skill, not a talent

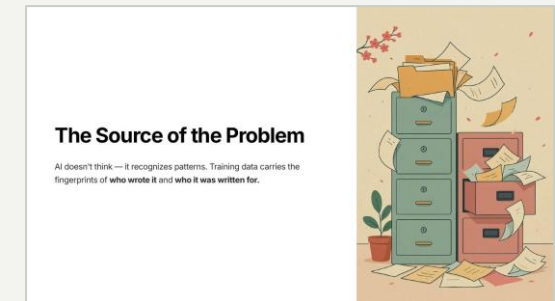
Four questions catch biased marketing copy before it costs you customers

Who is centred, whose perspective is absent, what is assumed, and the name of the pattern; then rewrite.

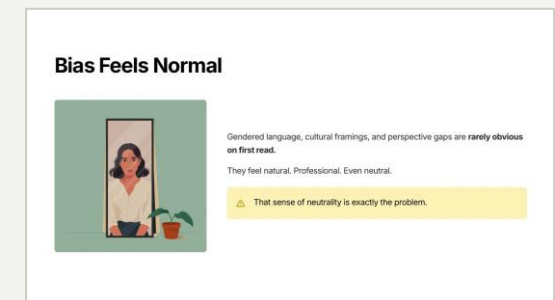
From video V16



V16, page 1 of 20. Spotting and fixing bias in AI marketing copy



V16, page 2 of 20. The source of the problem



V16, page 3 of 20. Bias feels normal

Same product, two sentences. One speaks to about a third of the list.

The AI draft

“Our premium wealth management service helps successful businessmen and their wives build a legacy.” Of 10,000 prospects on the mailing list, about 3,500 fit that picture.

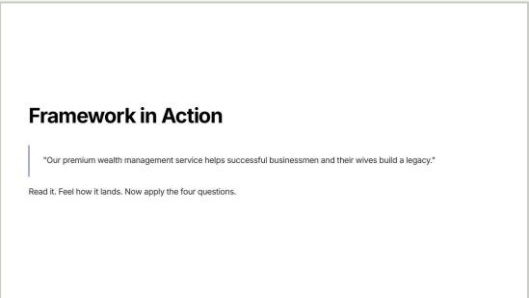
the word: exclusion

The rewrite

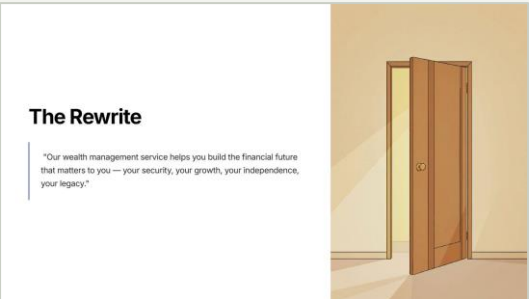
“Our premium wealth management service helps you build the financial future that matters to you: your security, your growth, your independence, your legacy.” About 9,500 can see themselves.

the word: inclusive language

The four questions find the problem in minutes, and the rewrite keeps the product, the tone and the word “premium”. At draft stage the fix costs minutes; after publication it costs customers.



V16, page 13 of 20. Framework in action



V16, page 16 of 20. The rewrite



V16, page 17 of 20. What changed

A story: Kodak's "Shirley cards", mid-1950s to mid-1990s

1



1950s: a card named Shirley

From the mid-1950s Kodak sent photo labs a reference print of a woman, known as a Shirley card. Technicians used it to set skin tones, shadows and light when printing customers' photos.

2



One face as the standard

The first card showed Shirley Page, a white woman who modelled for Kodak. Colour film was tuned for light skin, so darker skin often printed poorly, and mixed groups came out partly over- or underexposed.

3



Furniture and chocolate complain

In the 1960s and 1970s furniture and chocolate makers complained that dark and light woods, or dark and milk chocolate, looked the same in photos. A Kodak research manager later admitted dark skin had never been treated as a serious problem.

4



Better film, then a new card

Kodak improved its film. A later one, Kodak Gold Max, was advertised as able to photograph "the details of a dark horse in low light", a line that never mentions skin. A multiracial card followed in the mid-1990s.

5



Name the default, then fix it

One face was treated as normal, and every other customer was measured against it. AI copy can do the same with a default reader. Ask who the text assumes; once the pattern has a name, it can be fixed.

Sources: M. del Barco, "How Kodak's Shirley cards set photography's skin-tone standard", NPR, 13 November 2014; S. Lewis, "The racial bias built into photography", New York Times, 25 April 2019; L. Roth, "Looking at Shirley, the ultimate norm", Canadian Journal of Communication, 2009 (abstract); 99% Invisible, "Shirley cards"; Aperture, 2013 (checked 6 October 2026); retold in our own words.

If you remember nothing else

Three things to keep

- 1 Biased copy feels normal: it matches what the model saw most often, so it reads as neutral.
- 2 Four questions: who is centred, whose perspective is absent, what assumptions are encoded, and the name of the pattern (exclusion, misrepresentation, invisible frame).
- 3 Three risks: customers who leave, compliance review of regulated documents, and framing that joins your brand to a pattern you never chose.

From video V16

The Four-Question Checklist			
1	2	3	4
Who is centred? Who is the copy implicitly written for?	Who is absent? What groups or viewpoints are invisible?	What assumptions? Tone, formality, gendered language, cultural frames	Name the pattern Exclusion, misrepresentation, or invisible frame?

V16, page 19 of 20. The four-question checklist



Three Risks You Can't Ignore

Marketing Copy Speaks only to one demographic — others leave	Legal & Financial Docs Exclusionary language triggers compliance review
Strategic Messaging Reinforces power imbalances you didn't intend	

V16, page 4 of 20. Three risks you can't ignore

Apply Q2, “whose perspective is absent?”, to the bank’s sentence: “...helps successful businessmen and their wives build a legacy.” Who is absent?

- A Nobody: the sentence is polite and professional.
- B Single investors, women investing on their own account, same-sex couples, anyone outside one family model: on our list, about 6,500 of 10,000 prospects.
- C Only young people.
- D Only people who do not speak English.

From video V16

Q1 & Q2 Applied

Who is centred?

A male breadwinner and his spouse — one assumed family structure drives the entire sentence.

Who is absent?

Single investors, LGBTQ+ couples, female entrepreneurs, non-binary clients — **actively told this product isn't for them.**

V16, page 14 of 20. Q1 and Q2 applied

Q2: Whose Perspective Is Absent?



Being left out isn't neutral — It sends a signal.

☐ Absence is a form of exclusion, even when it's unintentional.

V16, page 10 of 20. Q2: whose perspective is absent?

Type the letter in the Zoom chat. The answer is discussed in the session.

An AI-drafted brochure for a credit union shows women only as “spouses” and “mothers” in its examples, never as account holders or business owners. Which pattern do you name?

- A Exclusion.
- B Misrepresentation: the group appears, but only in narrow roles.
- C Invisible frame.
- D No pattern: those roles are real.

Q4: Name the Pattern

 Exclusion Certain groups left out entirely	 Misrepresentation Groups appear in narrow or stereotyped ways	 Invisible Frame Assumptions you never chose — the model defaulted to them
---	--	--

Naming it makes it **fixable**.

V16, page 12 of 20. Q4: name the pattern

Risk 3: Skewed Framing



When AI consistently positions **one group as the actor** and another as the backdrop, your brand joins a broader pattern — one you probably didn't intend.

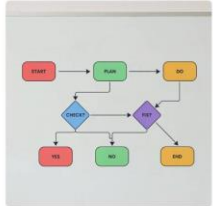
V16, page 7 of 20. Risk 3: skewed framing

Type the letter in the Zoom chat. The answer is discussed in the session.

Your fintech's AI-drafted loan page says: "Ideal for the family man planning ahead." Legal review is in two weeks; the page goes live tomorrow. What do you do?

- A Publish: legal will catch it later.
- B Run the four questions now and rewrite before publishing: minutes now, against lost customers, trust and a possible compliance review later.
- C Add a line saying the page was written with AI.
- D Ask the AI whether the sentence is biased.

Build It Into Your Workflow



Bias is **systematic** — it shows up consistently and quietly.

- Catch it at draft stage: costs minutes
- Catch it after publication: costs reputation, trust, compliance

☐ The earlier you apply the checklist, the less risk you carry.

V16, page 18 of 20. Build it into your workflow

Risk 2: Compliance Exposure



Culturally loaded or gender-specific terms in regulated documents can:

- Trigger compliance review
- Attract regulatory scrutiny
- Damage your reputation publicly

V16, page 6 of 20. Risk 2: compliance exposure

Type the letter in the Zoom chat. The answer is discussed in the session.

Your turn, described: four questions, then the money

The numbers are this block's mailing: the sentence sent to 10,000 prospects, a first-year fee of \$1,000 per new client. Tick each question as you apply it; Q2 and Q3 each unlock a fix. The panel shows the sentence as it now reads and what it brings in.

What you can change

Q1 Who is centred?

On or off; starts off. A married businessman. Naming him changes no word; it shows where to look.

Q2 Whose perspective is absent?

On or off; starts off. Single investors, women investing alone, same-sex couples. Unlocks the fix: "businessmen and their wives" becomes "you".

Q3 What assumptions are encoded?

On or off; starts off. Marriage, one family model, one goal. Fix: a legacy becomes security, growth, independence, legacy.

Q4 Name the pattern

On or off; starts off. Labels it exclusion. No number moves; the name is what lets the team brief the fix and spot it next time.

Where the sentence runs

Choose: A public web page / A regulated document (a fund prospectus, a client agreement). In regulated documents, wording that singles out one group can trigger a compliance review.

Try, in this order

1. Tick Q1 alone: no number moves; it shows where to look. Tick Q2: "you" replaces "successful businessmen and their wives", 8,500 prospects see themselves, 176 new clients instead of 96.
2. Tick Q3: four goals instead of one, 9,500 prospects and 192 clients: \$192,000 in first-year fees against \$96,000 at the start.
3. Untick everything and choose the regulated document: compliance review flags it. Tick Q1 and Q2: it passes.

The point. The four questions take minutes at draft stage. Skipped, the same sentence halves the result of the mailing and, in a regulated document, invites a review. Reply rates (2% of the prospects addressed, 0.4% of the others) and the shares are fixed teaching values; the arithmetic is count times rate times fee.

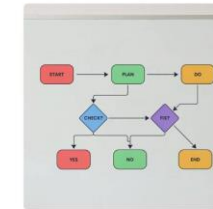
Before we move on

Three things from this block

- 1 Run the four questions on every AI draft before approval, not after the complaints.
- 2 Rewrite to the reader's goals, not to one assumed household; keep the product and the tone.
- 3 Count what exclusion costs: in our mailing, 96 clients against 192.

From video V16

Build It Into Your Workflow



Bias is **systematic** — it shows up consistently and quietly.

- Catch it at draft stage: costs minutes
- Catch it after publication: costs reputation, trust, compliance

☐ The earlier you apply the checklist, the less risk you carry.

V16, page 18 of 20. Build it into your workflow



Coming Up Next

Instead of catching bias at the end — **build prompts that reduce it from the start.**

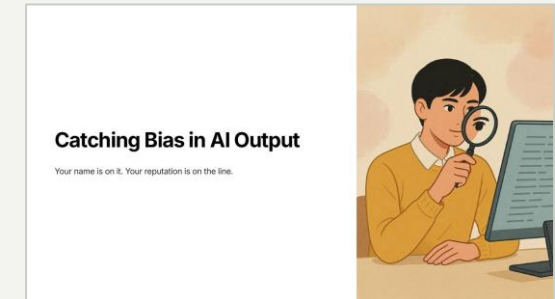
Learn how to brief an AI model so it's less likely to exclude or misrepresent, before the first word is generated.

V16, page 20 of 20. Coming up next

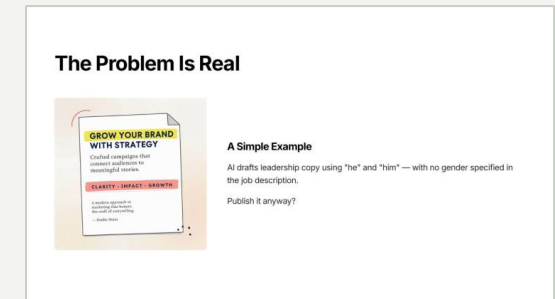
Prompts, careful reading and reviewers reduce bias; none removes it, and you stay accountable

Three ways to catch it, three ways to reduce it before output, a human reviewer before publication, and a clear disclosure line.

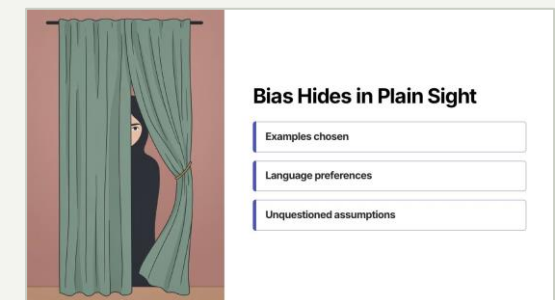
From video V17



V17, page 1 of 20. Catching bias in AI output



V17, page 2 of 20. The problem is real



V17, page 3 of 20. Bias hides in plain sight

Each safeguard catches part of what is left. None gets you to zero.

One safeguard

Only a reviewer at the end. Of 200 drafts in a quarter, 48 carry a biased passage: the reviewer catches 32 and 16 go out. About 20 hours of review.

the word: human in the loop

Five safeguards in a row

A specific prompt, variation testing, careful reading, several framings, then the reviewer: 4 of the 48 go out. About 53 hours a quarter.

the word: mitigation

The floor never reaches zero, because the model's weights do not change while you use it. Whatever gets through is still yours: you published it.

Mitigation Reduces Risk — Not Eliminates It

 **Specific prompts**
Take time to craft

 **Diverse feedback**
Costs resources

 **Multiple framings**
Extra work before publishing

Every strategy carries tradeoffs. Decide what makes sense for your stakes.

V17, page 15 of 20. Mitigation reduces risk — not eliminates it

The floor never reaches zero.

AI model weights are frozen at inference time. Internal biases are baked in — you can't retrain on the fly.

V17, page 14 of 20. The floor never reaches zero

You remain accountable. Full stop.

If bias slips through, that's on you. The AI didn't publish it — you did.

V17, page 18 of 20. You remain accountable. Full stop.

A story: the health app that forgot a cycle, 2014 to 2015

1



2014: your health in one place

On 2 June 2014 Apple announced a new Health app. It would gather the health and fitness data you chose into what Apple called “a clear and current overview in one place”.

2



It could log molybdenum

The app arrived with iOS 8 on 17 September 2014. Among the things it could record were how much molybdenum, a trace mineral, you had eaten, and your blood alcohol level.

3



But not a menstrual cycle

There was no way to record a menstrual cycle. Apple’s health data system had no category for it until the next major version of the software.

4



2015: reproductive health added

In June 2015 Apple announced reproductive-health data, including menstrual flow, for iOS 9. It shipped on 16 September 2015, a year after the first version.

5



Ask the people it is about

Anyone who tracks a cycle would likely have spotted the gap at a glance. Show drafts to the people they describe before you publish; they catch what authors miss. What still slips through is yours.

Sources: Apple Newsroom, “Apple unveils iOS 8” (2 June 2014) and “Apple previews iOS 9” (8 June 2015); Apple Developer, HealthKit documentation and “What’s new in iOS 9”; Wikipedia, “iOS 8”, “iOS 9”, “Health (Apple)” (checked 6 October 2026); retold in our own words.

If you remember nothing else

Three things to keep

- 1 Catch it: vary the prompt and compare, read for default assumptions, ask people from the communities the text describes.
- 2 Reduce it before output: name the audience, name the perspective, ask for several framings; then a human reviewer before publication.
- 3 None reaches zero: the weights are frozen while you use the model. Disclose clearly; you remain accountable.

From video V17

Your Bias Mitigation Checklist

01	02	03
Prompt-variation testing	Read for default assumptions	Seek representation feedback
04	05	
Human review before publication	Disclose AI involvement clearly	

V17, page 20 of 20. Your bias mitigation checklist

Mitigation Reduces Risk — Not Eliminates It

 Specific prompts Take time to craft	 Diverse feedback Costs resources	 Multiple framings Extra work before publishing
---	--	--

Every strategy carries tradeoffs. Decide what makes sense for your stakes.

V17, page 15 of 20. Mitigation reduces risk — not eliminates it

An AI tool drafted a job advert for a team-lead role using “he” and “him” throughout; the brief never mentioned gender. Your manager says: “The AI wrote it, not us.” If it is published, who is accountable?

- A The AI vendor.
- B Your organisation and you: whoever publishes the text owns it.
- C Nobody: it was automated.
- D The people whose writing trained the model.

Type the letter in the Zoom chat. The answer is discussed in the session.

The Problem Is Real



A Simple Example

AI drafts leadership copy using “he” and “him” — with no gender specified in the job description.

Publish it anyway?

V17, page 2 of 20. The problem is real

You own what's in there.

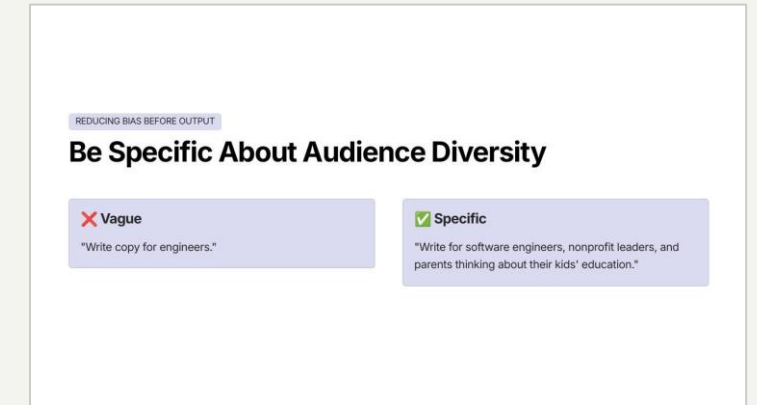
The AI didn't publish it — you did. Biased patterns and all.

V17, page 4 of 20. You own what's in there

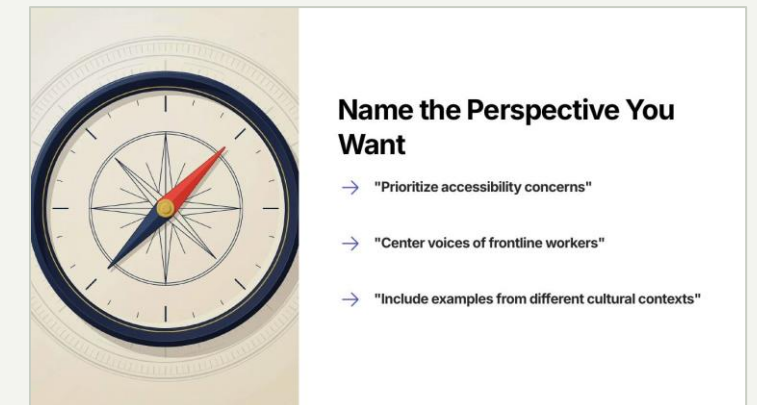
Which of these reduces bias before the model writes anything?

- A A human reviewer with lived experience of the subject.
- B Naming the audience and the perspective in the prompt: “for software engineers, nonprofit leaders and parents; prioritise accessibility”.
- C A disclosure line at the bottom of the text.
- D A correction after publication.

From video V17



V17, page 9 of 20. Be specific about audience diversity



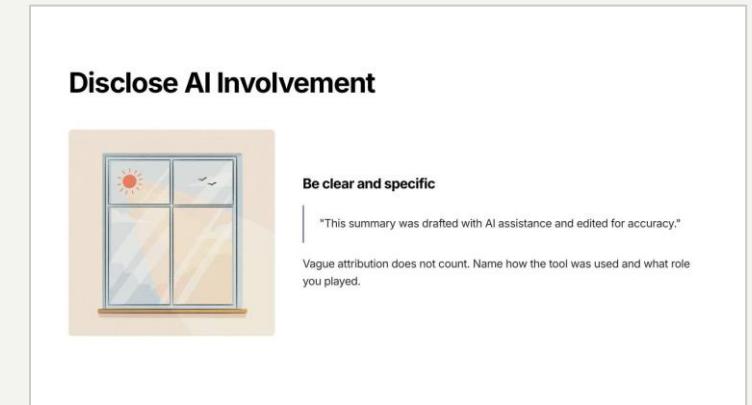
V17, page 10 of 20. Name the perspective you want

Type the letter in the Zoom chat. The answer is discussed in the session.

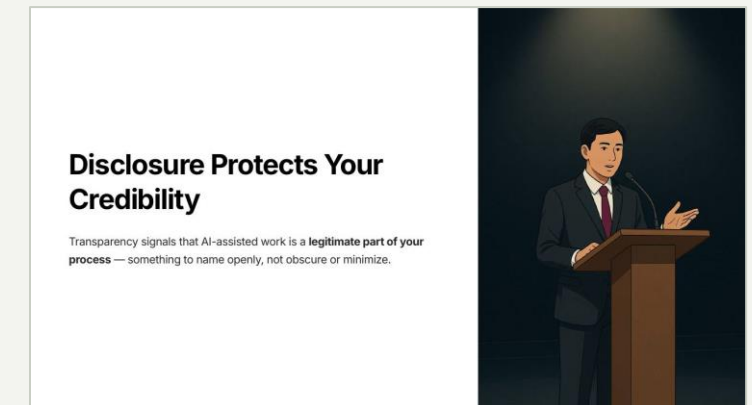
Which disclosure line meets V17's standard of clear and specific?

- A "Powered by AI."
- B "This summary was drafted with AI assistance and edited for accuracy by our communications team."
- C "Some content may have been generated."
- D No line: a person edited the text, so no disclosure is needed.

From video V17



V17, page 16 of 20. Disclose AI involvement



V17, page 17 of 20. Disclosure protects your credibility

Type the letter in the Zoom chat. The answer is discussed in the session.

Your turn, described: stack the safeguards

The numbers are this block's quarter: 200 AI-drafted pieces, 48 with a biased passage, each read by about 2,000 people. Switch safeguards on and off, in the order they happen, and choose when the reviewer reads. The panel follows the 48 passages and counts the hours.

What you can change

Name the audience and the perspective in the prompt

On or off; starts off. Prevention: about 1 biased passage in 4 is never written. About 1 minute per draft.

Prompt-variation testing: rephrase, rerun, compare

On or off; starts off. Catches about 1 in 3 of the passages still there. About 4 minutes per draft.

Read for default assumptions (the three questions)

On or off; starts off. Catches about 1 in 4 of what remains. About 2 minutes per draft.

Ask for two or three framings and choose

On or off; starts off. Catches about 1 in 3 of what remains. About 3 minutes per draft.

A human reviewer with lived experience

On or off; starts on. Catches about 2 in 3 of what remains. About 6 minutes per draft.

When the reviewer reads

Choose: Before publication / After publication. After publication, readers have already seen the passages; a correction reaches about 1 reader in 3.

Disclose AI involvement on each piece

On or off; starts off. Tells readers how the text was made. It catches nothing; it protects credibility.

Try, in this order

1. Reviewer only (the start): 32 caught, 16 go out, 20 hours. Tick all five safeguards: 4 go out, about 53 hours. Never zero.
2. All five on, move the reviewer to after publication: 12 passages go out first, 24,000 reader views, and the corrections reach about 5,333 readers.
3. Keep only the prompt and the reviewer (before): 12 go out, about 23 hours. Tick disclosure: no number moves.

The point. Mitigation lowers the count and costs time; the right stack depends on the stakes. Whatever is left, you published, so it is yours.

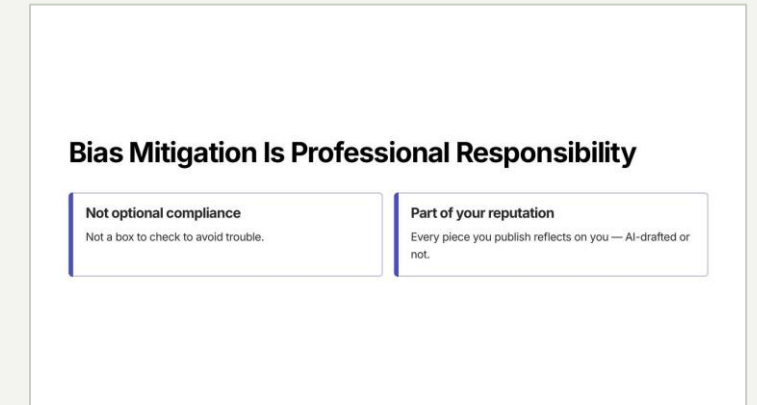
The catch rates, minutes and readers per piece are fixed teaching values chosen to show the trade-off; not measurements.

Before we move on

Three things from this block

- 1 Choose your stack of safeguards by the stakes, and write down which ones you used.
- 2 Review before publication: corrections rarely reach everyone who saw the original.
- 3 A disclosure line names the tool's part and yours.

From video V17



V17, page 19 of 20. Bias mitigation is professional responsibility

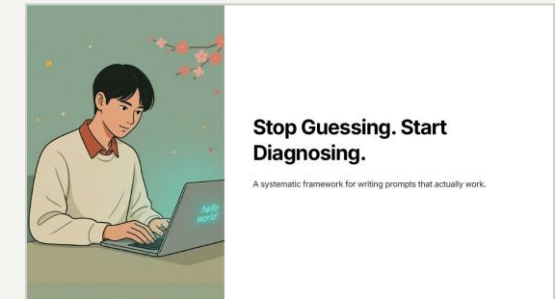


V17, page 13 of 20. Timing matters

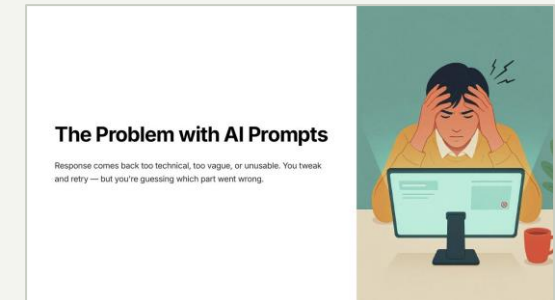
Every prompt has four parts: role, instruction, context and format

When the output fails, find the part that broke and fix only that part.

From video V18



V18, page 1 of 20. Stop guessing. Start diagnosing.



V18, page 2 of 20. The problem with AI prompts



V18, page 3 of 20. Effective prompt writers don't guess

“Write me something about our new product” against four filled slots.

One vague line

Only an instruction, and a vague one. The model guesses the voice, the angle, the facts and the shape: in our example, about 40 minutes of rework per prompt.

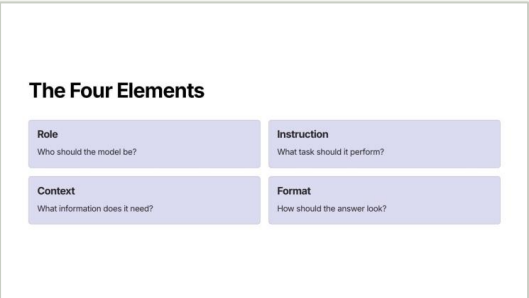
the word: instruction

Four slots

Role: marketing copywriter. Instruction: ad copy stressing speed and ease. Context: a cloud backup tool, two-minute set-up, \$15 a month; the competitor charges \$60. Format: a headline and two paragraphs.

the word: prompt template

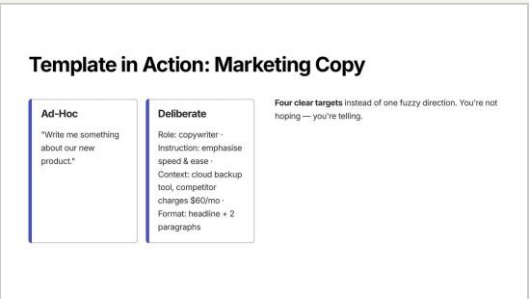
Four clear targets instead of one fuzzy direction. And when the output fails, you ask which of the four broke and fix only that one.



V18, page 4 of 20. The four elements



V18, page 14 of 20. The template



V18, page 15 of 20. Template in action: marketing copy

A story: the bomber that needed a checklist, 1935 to 1936

1



1935: a new bomber on trial

In 1935 the US Army was testing bombers from three makers at Wright Field in Dayton, Ohio. Boeing's entry was the Model 299, a large new plane first flown on 28 July 1935.

2



30 October: a crash on take-off

On 30 October 1935 the plane climbed, stalled and crashed soon after take-off. Two of the five men aboard died: the Army pilot, Major Ployer Hill, and Boeing's chief test pilot, Leslie Tower.

3



The cause: one step skipped

The gust lock, which holds the control surfaces still while the plane is parked, had not been released. On a new and complicated plane, one step had been missed.

4



13 planes and a checklist

The main contract went to another maker, but in January 1936 the Army ordered 13 of Boeing's planes for testing. Pilots wrote step-by-step checklists for each stage of flight. The plane became the B-17.

5



Fill every slot, find the break

A checklist names every step, so nothing rests on memory. A prompt works the same way: fill in role, instruction, context and format. When it fails, find the one slot that broke and fix only that.

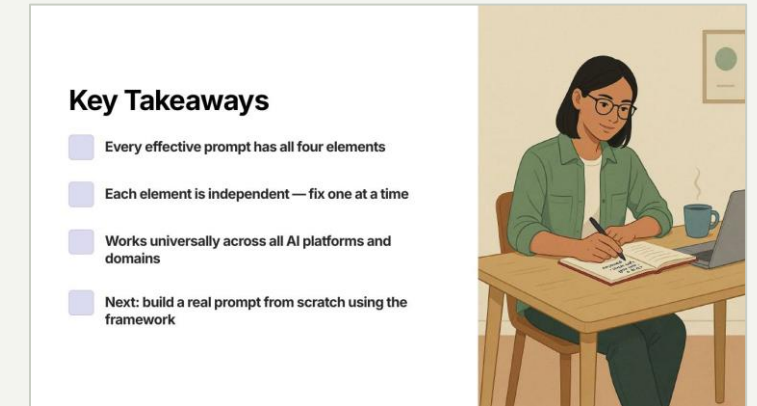
Sources: Wikipedia, "Boeing B-17 Flying Fortress" and "Ployer Peter Hill"; J. Schamel, "How the pilot's checklist came about" (checked 6 October 2026); retold in our own words.

If you remember nothing else

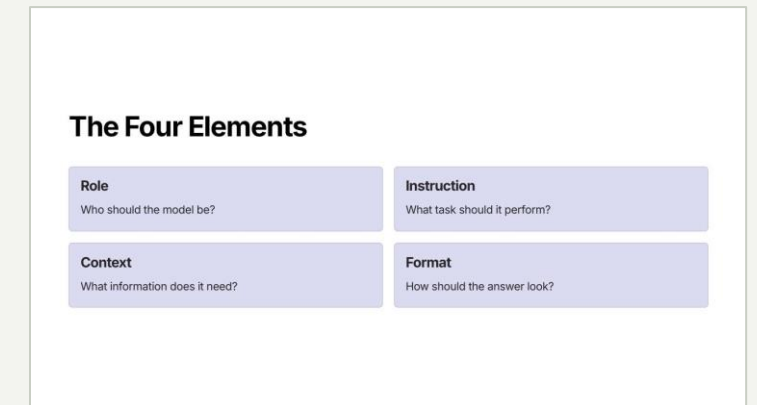
Three things to keep

- 1 Four elements: role (who the model should be), instruction (what task), context (what information), format (what shape).
- 2 They are independent: change one and the other three stay as they were.
- 3 Missing context is where invented facts come from; a missing format is where rework comes from.

From video V18



V18, page 20 of 20. Key takeaways



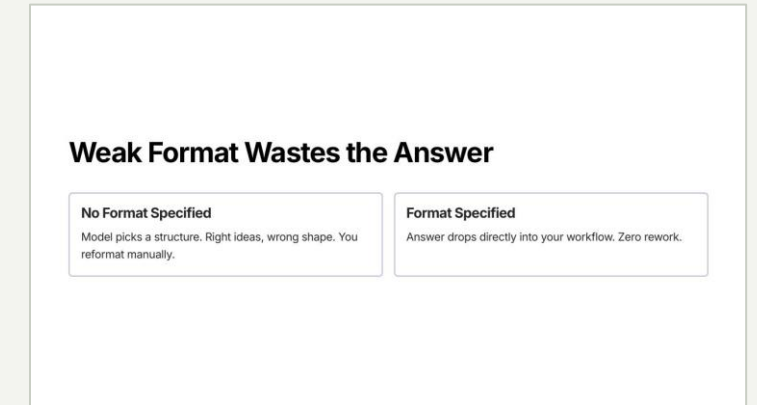
V18, page 4 of 20. The four elements

Question 1 of 3 · 45 seconds

The ad copy has the right ideas, but it arrives as one long block. The product page needs a headline and two short paragraphs, so you reformat it by hand every time. Which element is broken?

- A Role.
- B Instruction.
- C Context.
- D Format.

From video V18



V18, page 12 of 20. Weak format wastes the answer



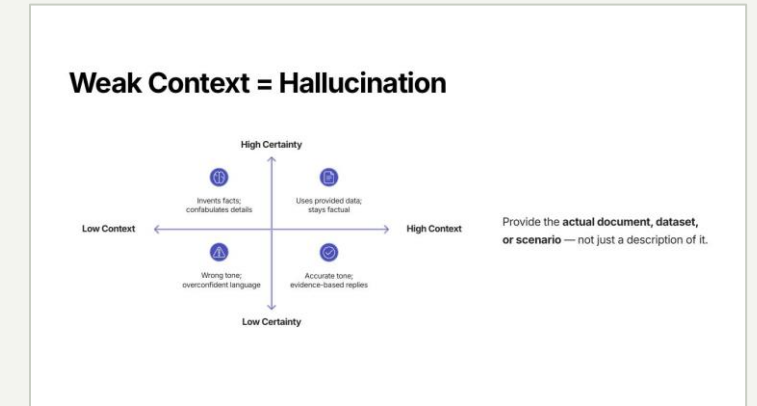
V18, page 11 of 20. Element 4: format

Type the letter in the Zoom chat. The answer is discussed in the session.

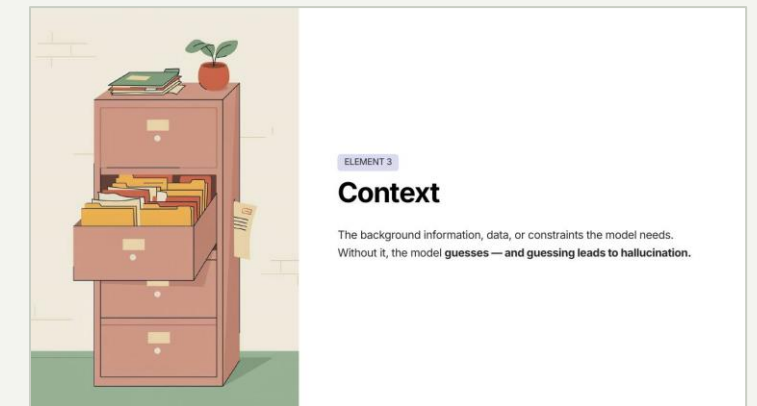
The copy says the backup tool costs “\$19 a month”. Nobody gave it a price; the real price is \$15. Which element was missing, and what does V18 call the result?

- A Role; a tone problem.
- B Instruction; a task problem.
- C Context; without the real figure the model filled the gap with a plausible one: a hallucination.
- D Format; a layout problem.

From video V18



V18, page 10 of 20. Weak context = hallucination



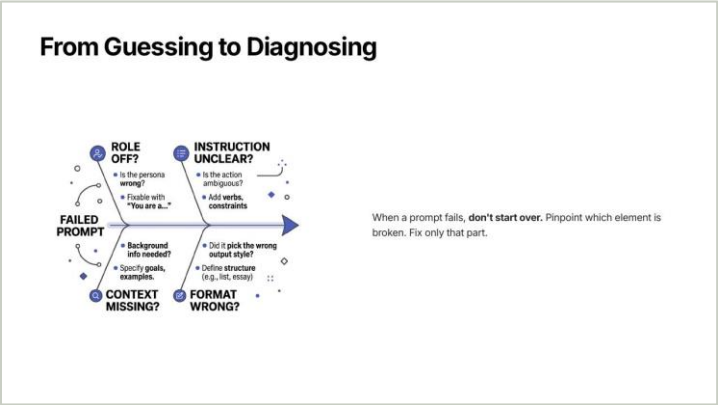
V18, page 9 of 20. Element 3: context

Type the letter in the Zoom chat. The answer is discussed in the session.

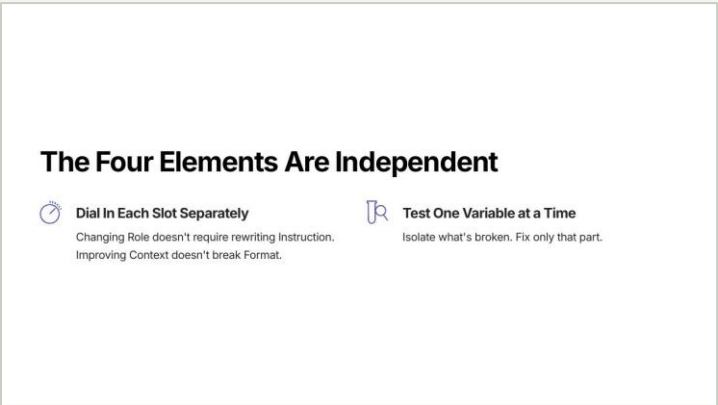
A prompt fails. A colleague rewrites all four elements at once and the new version works. What did the team lose?

- A Nothing: it works.
- B The diagnosis: nobody knows which element was broken, so the next failure starts from guessing again.
- C Money: four changes cost more than one.
- D The role.

From video V18



V18, page 18 of 20. From guessing to diagnosing



V18, page 13 of 20. The four elements are independent

Type the letter in the Zoom chat. The answer is discussed in the session.

Your turn, described: fill the four slots

The numbers are this block's job: ad copy for a cloud backup tool's product page, a headline and two paragraphs. Set the four slots. The panel shows what comes back, which slot causes each problem, and the rework per prompt and per month at 30 prompts.

What you can change

Role

Choose: No role / Marketing copywriter / Financial analyst / A five-year-old. Who the model should be. Sets the tone, the vocabulary and the depth.

Instruction

Choose: "Write something about our new product" / "Write ad copy that stresses speed and ease" / "Extract the three main claims and rank them". The task itself. A precise verb and goal decide what work gets done.

Context

Choose: No context / Only "the competitor charges \$60 a month" / Cloud backup, two-minute set-up, \$15 a month; competitor \$60. What only you know: the product, the numbers, the limits. Missing facts get invented.

Format

Choose: No format / A headline and two paragraphs / JSON with the fields headline and body. The shape of the answer. Name the shape the page needs and the text drops straight in.

Try, in this order

1. Start, nothing filled: 40 minutes of rework per prompt, 20 hours a month. Set the role to copywriter: only the voice line changes, 30 minutes.
2. Fill all four for the job (copywriter; ad copy on speed and ease; the full product facts; a headline and two paragraphs): 0 minutes.
3. Now set the context to "only the competitor charges \$60": one invented fact (your price) and 10 minutes come back. One slot, one symptom.

The point. Each slot controls one property of the answer. Because the slots are independent, a bad output points to the slot that broke, and the fix is one slot, not a new prompt.

Rework minutes are fixed teaching values chosen to show the pattern; not measurements.

Before we move on

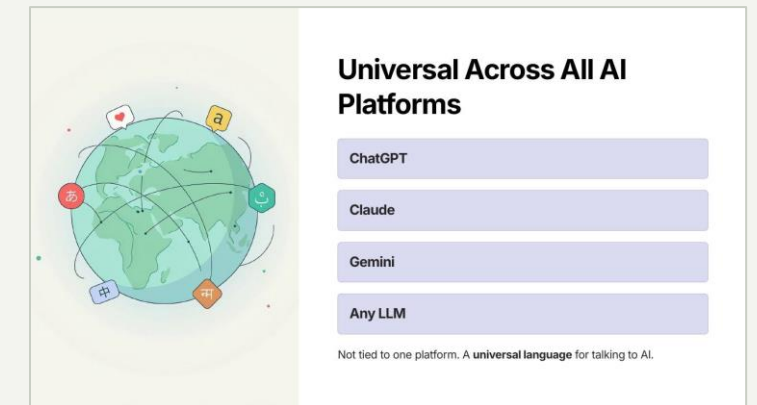
Three things from this block

- 1 Before you write, fill four slots; when the output fails, ask which slot broke.
- 2 The same four work in ChatGPT, Claude, Gemini or any other model, and in every field.
- 3 This is the vocabulary of Assignment 1, the Four-Variation Prompt Log.

From video V18



V18, page 19 of 20. Prompt writing as craft

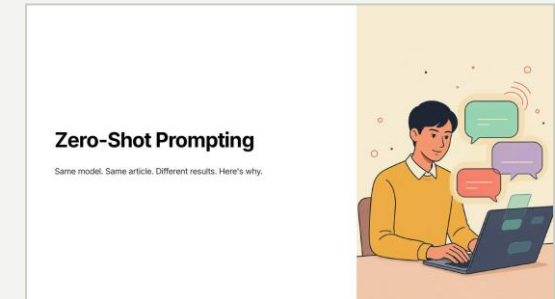


V18, page 16 of 20. Universal across all AI platforms

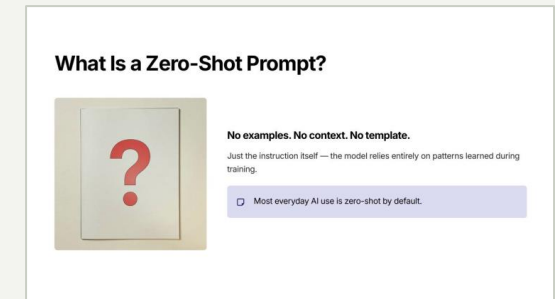
Run the plain prompt first and keep what it gives you: that is your baseline

Zero-shot means no examples. Its output shows the model's defaults and gaps, and every later change is measured against it.

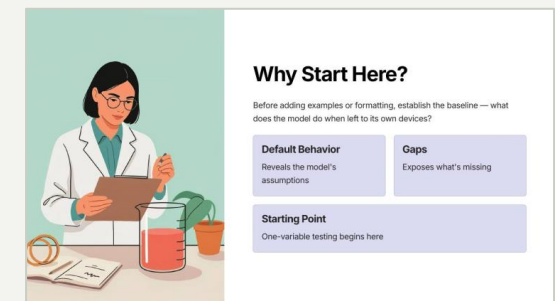
From video V19



V19, page 1 of 10. Zero-shot prompting



V19, page 2 of 10. What is a zero-shot prompt?



V19, page 3 of 10. Why start here?

“Explain machine learning.” Then one change: “...as if I’m a 10-year-old.”

The zero-shot baseline

A textbook paragraph: about 180 words, definitions, no example, a university reading level. It shows what the model does on its own.

the word: zero-shot

One change, measured

About 110 words in two short paragraphs, one everyday example, a 10-year-old’s vocabulary. You can say what moved because you kept the first answer.

the word: baseline

Without the recorded baseline, the second answer is just another answer. With it, the difference is a result.

What is a Zero-Shot Prompt?



No examples. No context. No template.

Just the instruction itself — the model relies entirely on patterns learned during training.

☒ Most everyday AI use is zero-shot by default.

V19, page 2 of 10. What is a zero-shot prompt?

A Real Example

“Explain machine learning.”

Zero-shot output: a generic, textbook-style paragraph — definitions, no depth.

What the baseline reveals:

- Model needs a defined audience
- Model needs a specified depth
- Model needs a target format

V19, page 5 of 10. A real example



Change One Variable

“Explain machine learning **as if I’m a 10-year-old.**”

Tone, vocabulary, and structure all shift — from a single small edit.

V19, page 6 of 10. Change one variable

A story: the first readings on Mauna Loa, 1958 to 2026

1



A chemist who measured carefully

In the mid-1950s a young chemist, Charles David Keeling, built a very precise way to measure carbon dioxide in air. Far from cities and forests he kept getting almost the same value: about 310 parts per million.

2



A station high on a volcano

With the Scripps Institution of Oceanography and the US Weather Bureau, an instrument went up on Mauna Loa, Hawaii, about 3,400 metres up and far from local pollution. On 29 March 1958 it read 313 parts per million.

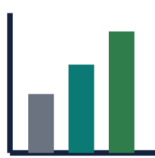
3



A yearly cycle, then a rise

Within a year the readings showed a cycle: a peak in May, a low in October while northern plants grew, then up again. The 1959 average was higher than 1958's, and 1960's higher again.

4



Keeping the record going

Funding was never secure, and budget cuts stopped the measurements for a while in 1964. The record, now called the Keeling Curve, still runs: in May 2026 the monthly average at Mauna Loa was 432.

5



Why the first reading matters

The rise from 313 to 432 can be measured only because the early readings were taken with care and kept. Your zero-shot output plays the same part: record the plain answer first, so every change has a reference.

Sources: Scripps Institution of Oceanography, “The history of the Keeling Curve”, “The early Keeling Curve” and Keeling biography; UC San Diego, “Annual carbon dioxide peak reaches 432 parts per million” (June 2026); NOAA, Mauna Loa record named a National Historic Chemical Landmark (checked 6 October 2026); retold in our own words.

If you remember nothing else

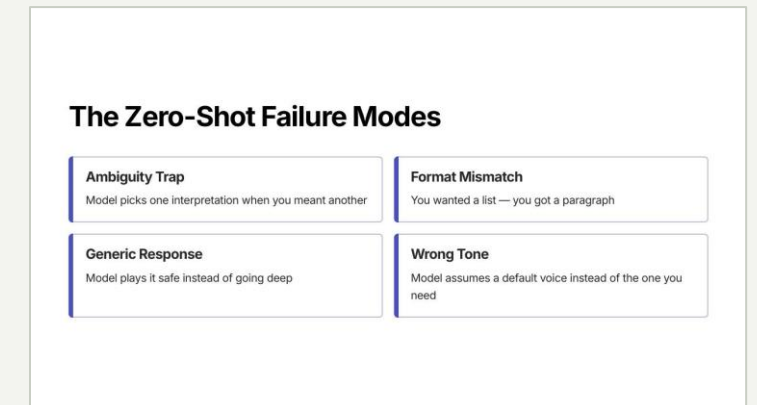
Three things to keep

- 1 Zero-shot: the instruction alone, with no examples; most everyday use is zero-shot without anyone noticing.
- 2 Its output shows the model's defaults and gaps: the ambiguity trap, format mismatch, a generic answer, the wrong tone.
- 3 Record the zero-shot output first, then change one thing and measure the difference.

From video V19



V19, page 9 of 10. Document your baselines

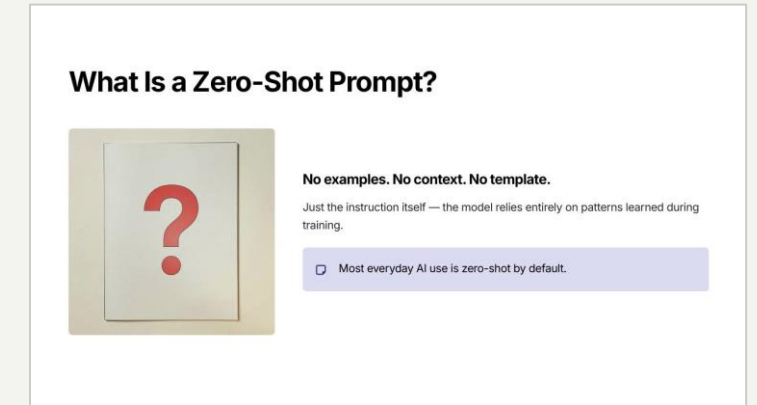


V19, page 4 of 10. The zero-shot failure modes

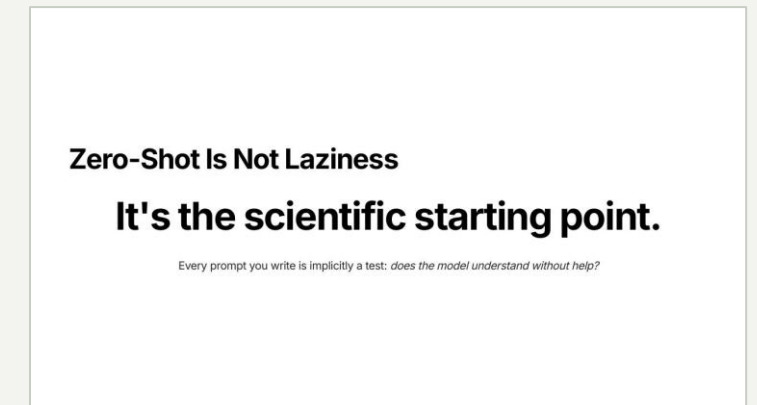
Which of these is a zero-shot prompt?

- A “Summarise this report. Here are two summaries we liked, as examples: ...”
- B “Summarise this report.” and nothing else.
- C A prompt that returned no answer.
- D A prompt with no context but three examples.

From video V19



V19, page 2 of 10. What is a zero-shot prompt?



V19, page 8 of 10. Zero-shot is not laziness

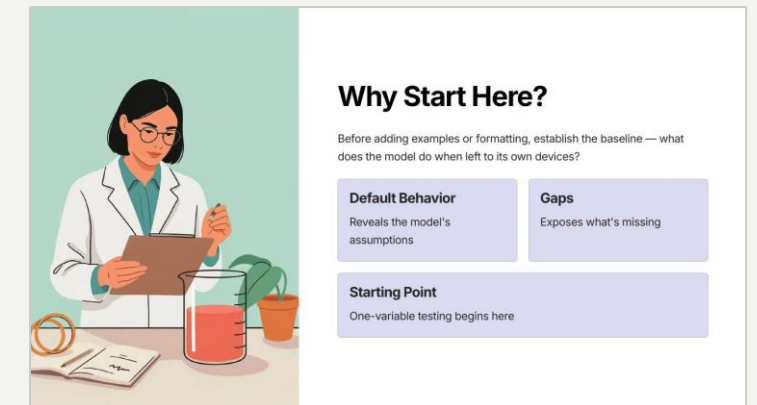
Type the letter in the Zoom chat. The answer is discussed in the session.

Before improving a prompt for client summaries, why does V19 ask you to save the zero-shot output first?

- A To prove that the AI is weak.
- B It is the baseline: the reference every later change is measured against, so you can say what each change did.
- C Because regulators require it.
- D Because the zero-shot output is always the best one.



V19, page 9 of 10. Document your baselines

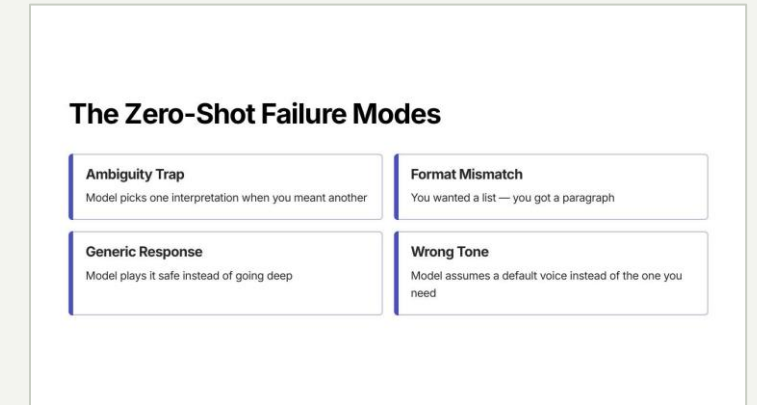


V19, page 3 of 10. Why start here?

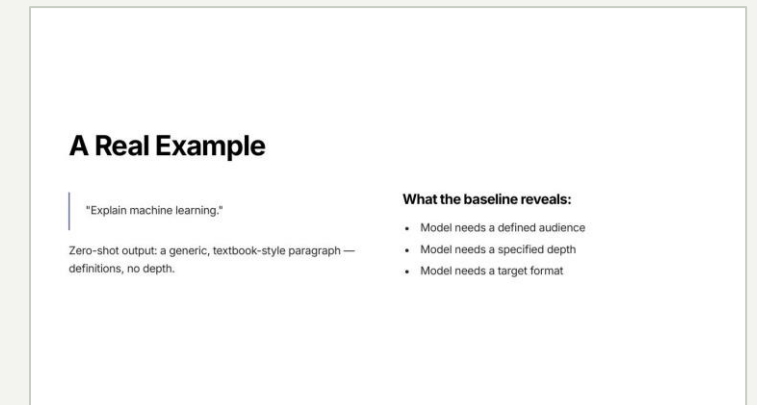
Type the letter in the Zoom chat. The answer is discussed in the session.

Two colleagues ran “Summarise this report.” One got bullet points, the other a paragraph. You needed bullets for a slide. Which zero-shot failure mode is this?

- A The ambiguity trap.
- B Format mismatch.
- C A generic response.
- D The wrong tone.



V19, page 4 of 10. The zero-shot failure modes



V19, page 5 of 10. A real example

Type the letter in the Zoom chat. The answer is discussed in the session.

Your turn, described: baseline first, then one change

The numbers are this block's prompt, "Explain machine learning.", run zero-shot. Choose the job, the one thing you add, and whether you kept the zero-shot output. The panel describes the answer, lists the failure modes left for your job, and says whether the change can be measured.

What you can change

The job you need the answer for

Choose: A short, friendly paragraph for parents at a school evening / Three bullet points for a client slide / A note for a bank's risk committee. What the answer is for. The same output can fit one job and fail another.

What you add to "Explain machine learning."

Choose: Nothing: the zero-shot prompt / "...as if I'm a 10-year-old." / "...in three bullet points." / "...in under 50 words." / "...for a bank's risk committee.". The one variable you add. One addition at a time, so the change can be traced to it.

I kept the zero-shot output (date, model, full text)

On or off; starts on. The baseline. Without it you can describe a new answer but not measure what your change did.

Try, in this order

1. Job: parents' evening, nothing added: all 4 failure modes are present. This is the baseline: record it.
2. Add "...as if I'm a 10-year-old": 0 of 4 for the parents' job, and the panel can say what moved: 180 words became 110, one example appeared.
3. Keep that addition and switch the job to the client slide: 2 of 4 fail (format and tone). Untick "I kept the zero-shot output": the change can no longer be measured.

The point. Zero-shot is not lazy: it is the control in your experiment. Keep it, change one thing, and the difference becomes something you can name.

Word counts and failure modes are fixed teaching values describing typical outputs; your tool's will differ, which is why you keep your own baseline.

Three things from this block

- 1 Your first run of any new prompt is an experiment; keep its result.
- 2 Add one constraint at a time to see which constraint mattered.
- 3 Few-shot prompting, adding worked examples, has no deck in this package; when you try it, compare it with your zero-shot baseline.

Zero-Shot Is Not Laziness

It's the scientific starting point.

Every prompt you write is implicitly a test: *does the model understand without help?*

V19, page 8 of 10. Zero-shot is not laziness

Up Next: Few-Shot Prompting

Give the model one or more examples before the task — then compare the output to your zero-shot baseline.

☐ Your documented zero-shot result is the comparison that makes few-shot testing meaningful.

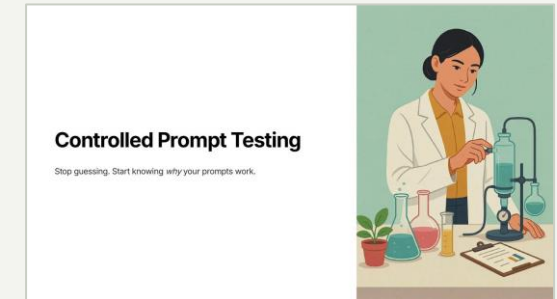


V19, page 10 of 10. Up next: few-shot prompting

Change exactly one element, run both versions on the same input, write down what moved

Change everything at once and the output may improve, but nobody can say why, and nobody can repeat it.

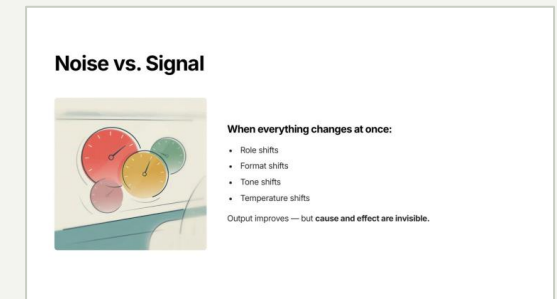
From video V20



V20, page 1 of 20. Controlled prompt testing



V20, page 2 of 20. The hidden cost of changing everything at once



V20, page 3 of 20. Noise vs. signal

Same instruction, context, format and tone. Only the role changes.

Version A: marketing copywriter

A note about a savings app. Warm and persuasive: 6 emotional words, 2 warm phrases, no figures, 1 call to action.

the word: role

Version B: financial analyst

The same note, the same input. Precise and measured: 0 emotional words, the same 2 warm phrases, 4 figures (return, fees, risk), no call to action.

the word: controlled test

Because nothing else changed, the difference belongs to the role. Written down in one sentence, it becomes evidence you can reuse.

A Concrete Example: One Prompt, Two Roles

Everything stays identical — instruction, context, format, tone. Only the role changes.

Version 1

Role: Marketing Copywriter

Version 2

Role: Financial Analyst

V20, page 6 of 20. A concrete example: one prompt, two roles

The Role Effect in Action

Marketing Copywriter

- Emotion-driven language
- Warm & persuasive tone
- Clear call-to-action

Financial Analyst

- Data & risk focus
- ROI & measurable returns
- Precise, metric-driven language

☐ **Key insight:** Role fundamentally reshapes voice, priorities, and evidence type.

V20, page 7 of 20. The role effect in action

Step 2: Change Exactly One Element



Write down **precisely** what you changed.

Keep the other four constant — no small tweaks, no cleanup edits.

⚠ Any extra changes reintroduce the noise you're trying to eliminate.

V20, page 10 of 20. Step 2: change exactly one element

A story: twelve sailors and one difference, 1747 to 1795

1



Scurvy on HMS Salisbury

In 1747, during ten weeks at sea, 80 of the 350 sailors on HMS Salisbury fell ill with scurvy, a disease caused by too little vitamin C. The ship's surgeon, James Lind, set out to compare the usual remedies.

2



Twelve sailors, kept alike

On 20 May 1747 he chose twelve sick sailors whose cases were, in his words, “as similar as I could have them”. They lay in the same quarters and ate the same diet. Only the remedy was meant to differ.

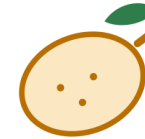
3



Six remedies, two men each

Each pair got one remedy: cider, elixir of vitriol (a dilute acid), vinegar, sea water, two oranges and a lemon a day, or a paste of garlic, mustard seed and myrrh.

4



The fruit pair recovered first

The fruit ran out after six days, yet one of that pair was soon fit for duty and the other well enough to nurse the rest. Lind published in 1753; the navy ordered lemon juice for its ships only in 1795.

5



Change one thing, write it down

Lind kept food and quarters the same, changed only the remedy and wrote down what happened. A controlled prompt test follows the same steps: keep everything identical, change one element, compare, record.

Sources: James Lind, A Treatise of the Scurvy (1753), the trial passage; James Lind Library, I. Milne, “Who was James Lind, and what exactly did he achieve?”; Royal College of Physicians of Edinburgh, “Scurvy in the College collections”; NHS, “Scurvy” (checked 6 October 2026); retold in our own words.

If you remember nothing else

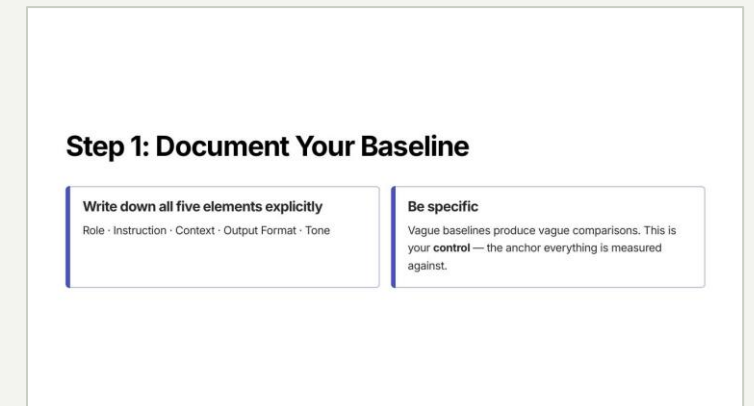
Three things to keep

- 1 Change exactly one element: role, instruction, context, format or tone (V20 counts tone as a fifth dial).
- 2 Run both versions on identical input, compare them side by side, write one or two sentences on what moved.
- 3 Then repeat with the next element; each valid test adds one line to your evidence base.

From video V20



V20, page 8 of 20. The 5-step framework



V20, page 9 of 20. Step 1: document your baseline

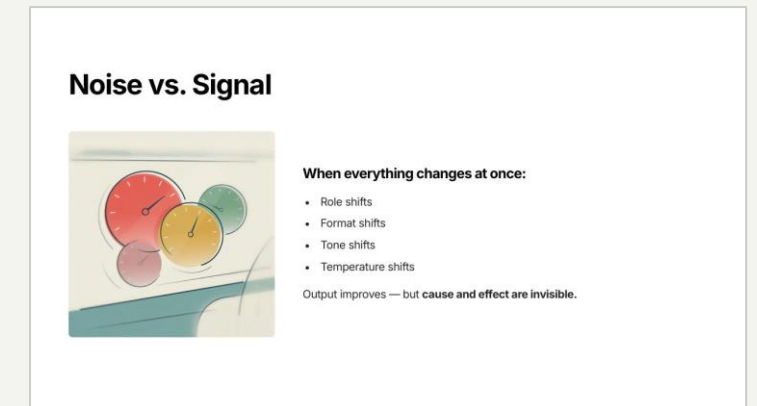
Your team changed the role, the tone and the instructions of a client-email prompt at the same time. The emails got better. What is the problem?

- A None: better is better.
- B Nobody can say which change helped, so the improvement cannot be repeated or explained.
- C Three changes take too long.
- D The temperature must have changed too.

From video V20



V20, page 2 of 20. The hidden cost of changing everything at once



V20, page 3 of 20. Noise vs. signal

Type the letter in the Zoom chat. The answer is discussed in the session.

To compare two roles, a colleague runs version A on Monday's customer complaint and version B on Tuesday's. Is this a valid one-variable test?

- A Yes: the role is the only change in the prompt.
- B No: the input changed too, which is a second variable.
- C Yes, as long as both runs use the same model.
- D No, because complaints are confidential.

Step 3: Run Both Versions on Identical Input



If input varies even slightly, you've introduced a second variable — the comparison loses its meaning.

V20, page 11 of 20. Step 3: run both versions on identical input

Step 2: Change Exactly One Element



Write down **precisely** what you changed.

Keep the other four constant — no small tweaks, no cleanup edits.

⚠ Any extra changes reintroduce the noise you're trying to eliminate.

V20, page 10 of 20. Step 2: change exactly one element

Type the letter in the Zoom chat. The answer is discussed in the session.

Your valid test: copywriter against financial analyst, everything else identical. What do you write in your log?

- A “B is better.”
- B One or two sentences on what changed and what it did, for example “The role alone moved the text from emotion and a call to action to figures and risk.”
- C Nothing: the outputs speak for themselves.
- D Only the name of the model.

From video V20

Step 4: Compare Side-by-Side & Take Notes



Ask yourself:

- Did the tone change?
- Level of detail?
- Structure or reasoning style?

This reflection is where the experiment becomes knowledge.

V20, page 12 of 20. Step 4: compare side-by-side & take notes

Reflect — Don't Skip This Step

Write a sentence or two explaining what that element did. This is the step that turns a one-time experiment into something you can **repeat and build on**.

Which output better serves your goal? What specifically changed?

V20, page 17 of 20. Reflect — don't skip this step

Type the letter in the Zoom chat. The answer is discussed in the session.

Your turn, described: the one-variable test bench

The numbers are this block's prompt: a note about a savings app with five elements (V20 adds tone to V18's four). Version A is the baseline. Tick what changes in version B, then press "Log this test". The panel says whether the test is valid and what you may write down.

What you can change

Role: marketing copywriter → financial analyst

On or off; starts off. Role shapes the voice and the priorities.

Instruction: main benefit → three main risks

On or off; starts off. Instruction changes the task itself.

Context: add three customer reviews to the facts

On or off; starts off. Context changes the facts the model can use.

Format: one paragraph → a three-column table

On or off; starts off. Format changes the shape of the answer.

Tone: warm → neutral

On or off; starts off. Tone changes the emphasis, not the facts.

Run version B on a different customer question

On or off; starts off. Not a prompt element: the input. If it differs, that is one more variable.

Log this test (add it to the evidence base)

A button. Records the test only if exactly one thing changed; otherwise the log refuses it.

Try, in this order

1. Tick the role change: 1 variable, valid. Emotional words 6 → 0, figures 0 → 4, calls to action 1 → 0. Press "Log this test": 1 of 5 elements mapped.
2. Also tick the tone change: 2 variables, not valid, and the log refuses. Untick it.
3. With only the role changed, tick "different customer question": 2 variables again. Identical input is part of the method.

The point. One change, identical input, a written note: each valid test adds one line to what you know about each element, and that map is what lets you predict a prompt before you run it.

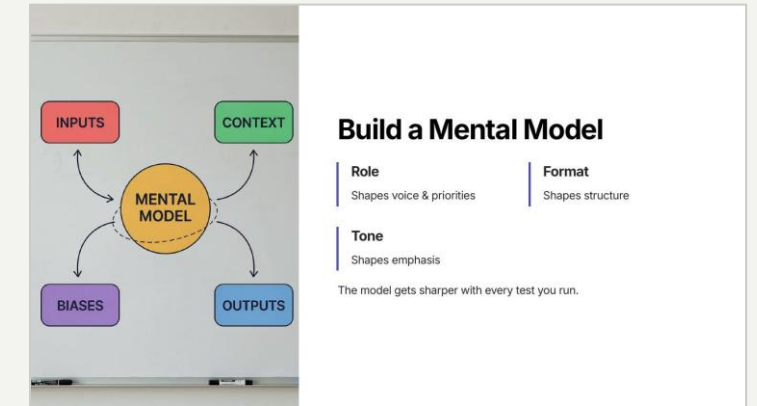
Counts are fixed teaching values describing typical outputs of the two roles; the live lab gives real ones.

Before we move on

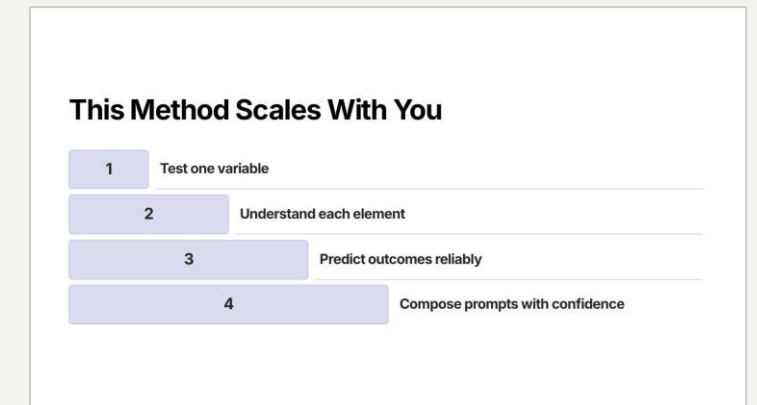
Three things from this block

- 1 Role shapes voice and priorities, format shapes structure, tone shapes emphasis.
- 2 This is Assignment 1: four variants, one element each, outputs in a table, due Monday 12 October.
- 3 Evidence beats intuition: you can predict a prompt you have tested.

From video V20



V20, page 18 of 20. Build a mental model



V20, page 19 of 20. This method scales with you

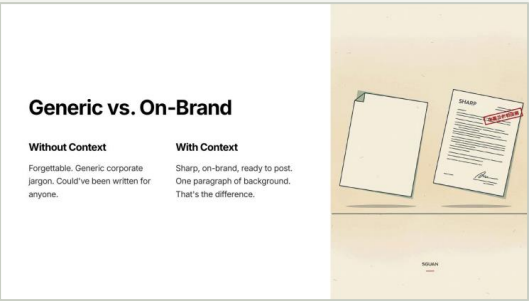
Context is what only you know; three to five dense sentences turn a generic answer into yours

The model has seen millions of product launches and never yours.

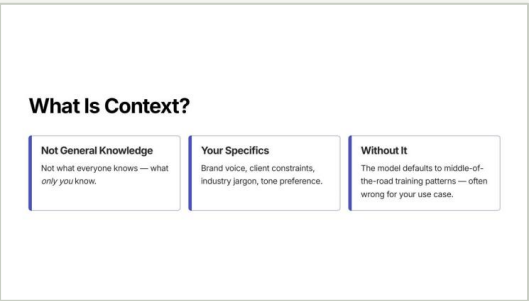
From video V21



V21, page 1 of 20. The power of context in prompt engineering



V21, page 2 of 20. Generic vs. on-brand



V21, page 3 of 20. What is context?

“Write a post about our new product launch.” The same model, with and without one paragraph.

Without context

Safe, corporate, could be any brand: the middle of everything the model has read. In our example it needs 4 rewrites before it can be posted.

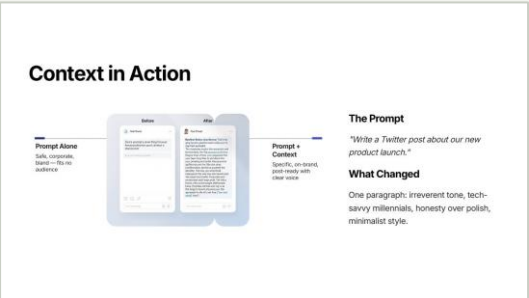
the word: training defaults

With a four-sentence context block

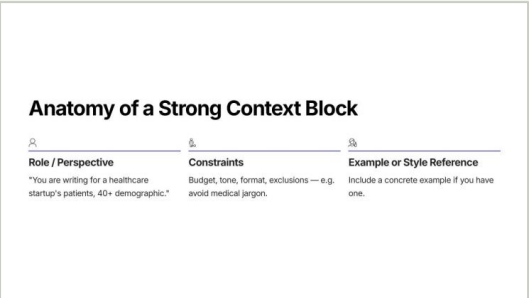
Who is writing, who reads it (tech-savvy millennials), how it should sound (irreverent, honest over polished), the limits (under 280 characters, minimalist). Eight facts; 0 rewrites.

the word: context

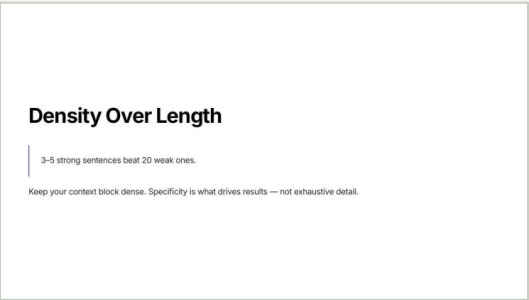
The model did not get smarter; it got a target. Density beats length: four sentences carrying eight usable facts beat twenty that carry the same eight.



V21, page 5 of 20. Context in action



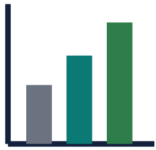
V21, page 7 of 20. Anatomy of a strong context block



V21, page 8 of 20. Density over length

A story: New Coke and its 79 days, 1985

1



Coca-Cola under pressure

By the early 1980s Coca-Cola had been losing market share for years, largely to Pepsi. The company decided to change the formula of its main drink for the first time in 99 years.

2



Nearly 200,000 taste tests

Nearly 200,000 people tasted a new, sweeter formula. In blind tests most preferred it to both the old Coke and Pepsi. About one tester in ten was angry at the very idea of replacing the original.

3



Launch day, then the backlash

On 23 April 1985 the new formula replaced the original, and many loyal drinkers were furious. By June the company's hotline was taking 1,500 calls a day, against about 400 before the change.

4



Back after 79 days

On 11 July 1985, 79 days after the launch, the company announced that the original was coming back as Coca-Cola Classic. In the two days after that announcement, its hotline took 31,600 calls.

5



Right answer, wrong question

The tests answered the question they asked: which drink tastes better? The company says they missed the bond people felt with their Coke. A prompt without your context gets the same kind of answer.

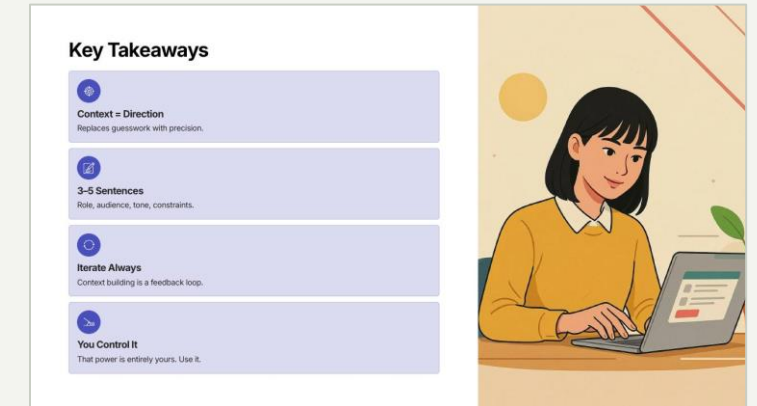
Sources: The Coca-Cola Company, "New Coke: the most memorable marketing blunder ever?"; History.com, "New Coke debuts, one of the biggest product flops in history"; Wikipedia, "New Coke" (checked 6 October 2026); retold in our own words.

If you remember nothing else

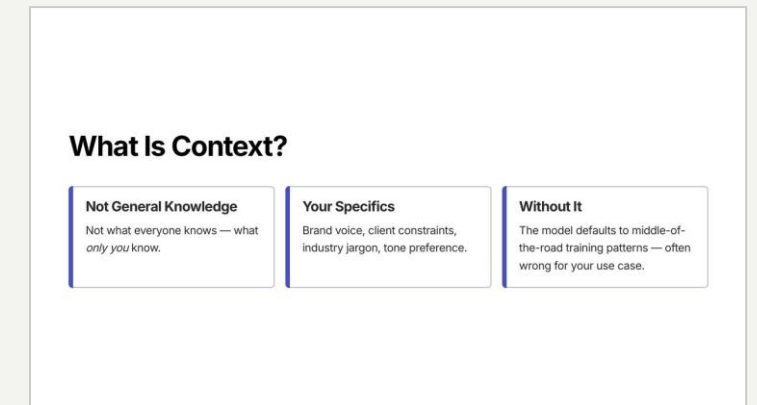
Three things to keep

- 1 Context is what the model cannot pull from its training: your brand, your audience, your constraints, your data.
- 2 A strong block: role or perspective, audience, tone, constraints, an example if you have one; three to five dense sentences.
- 3 Building context is a loop: run, read, add what was missing, run again.

From video V21



V21, page 20 of 20. Key takeaways

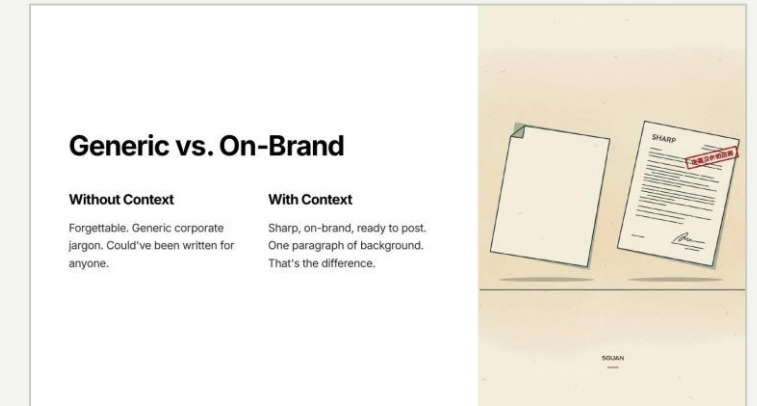


V21, page 3 of 20. What is context?

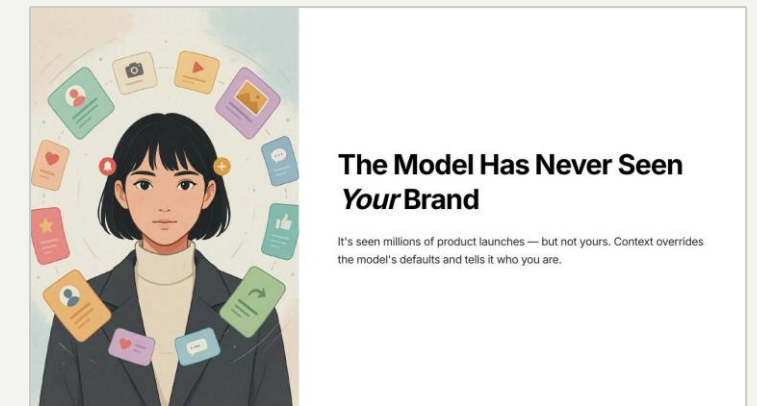
“Write a social media post about our new product launch.” The result is bland corporate text that could be about anyone. What is missing?

- A A bigger, more expensive model.
- B Context: who you are, who reads the post, how it should sound, and the limits.
- C A higher temperature.
- D More exclamation marks in the prompt.

From video V21



V21, page 2 of 20. Generic vs. on-brand



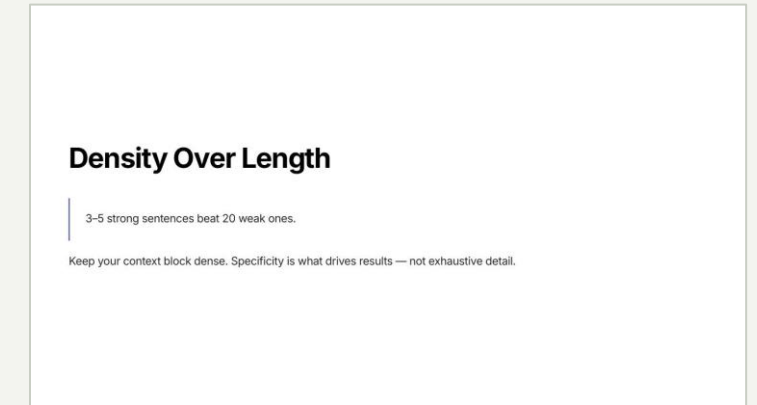
V21, page 4 of 20. The model has never seen your brand

Type the letter in the Zoom chat. The answer is discussed in the session.

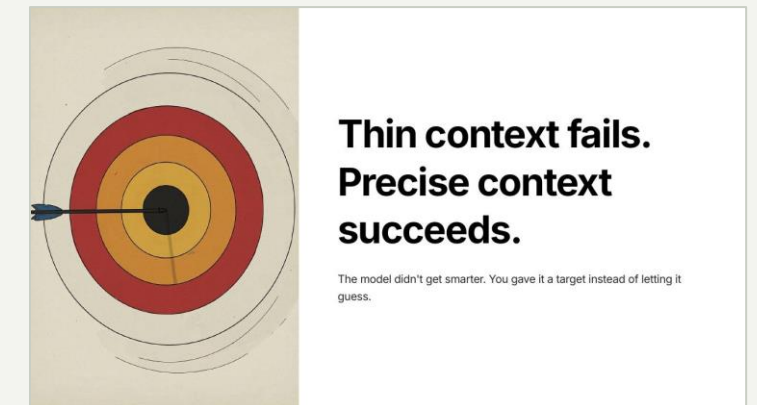
Two context blocks for the same post. A: twenty sentences of company history and mission statements. B: four sentences: role, audience, tone, constraints. Which works better, and why?

- A A: more information is always better.
- B B: dense, specific sentences carry the facts the post needs; filler buries them.
- C Neither: context does not change the output.
- D A, because the model can pick out what matters.

From video V21



V21, page 8 of 20. Density over length



V21, page 6 of 20. Thin context fails, precise context succeeds

Type the letter in the Zoom chat. The answer is discussed in the session.

A health start-up writes to patients over 40: a conversational email, a budget of fifty dollars, no medical jargon. In the context block, which part is “no medical jargon”?

- A Role or perspective.
- B Audience.
- C Constraint.
- D Example or style reference.

From video V21

Anatomy of a Strong Context Block

 Role / Perspective "You are writing for a healthcare startup's patients, 40+ demographic."	 Constraints Budget, tone, format, exclusions — e.g. avoid medical jargon.	 Example or Style Reference Include a concrete example if you have one.
---	--	---

V21, page 7 of 20. Anatomy of a strong context block

Your Context Block Template

Role
Who is writing, for whom?

Audience
Who will read this?

Tone
Playful? Professional? Conversational?

Constraints
Format, length, exclusions, budget.

V21, page 11 of 20. Your context block template

Type the letter in the Zoom chat. The answer is discussed in the session.

Your turn, described: build the context block

The numbers are this block's launch post. Tick the parts of the context block you write, one sentence each, and add filler sentences if you like. The panel counts sentences and usable facts and says how many rewrites the post needs before it goes out.

What you can change

Role: "You write for a start-up that sells a budgeting app."

On or off; starts off. Who is speaking. One sentence, two facts.

Audience: "Readers are tech-savvy millennials who distrust hype."

On or off; starts off. Who reads it. One sentence, two facts.

Tone: "Irreverent and playful; honest rather than polished."

On or off; starts off. How it should sound. One sentence, two facts.

Constraints: "Under 280 characters; minimalist style."

On or off; starts off. The limits: length, style, exclusions. One sentence, two facts.

Example: one of our past posts that worked, pasted in

On or off; starts off. A style reference. It lets the model copy a voice instead of guessing one.

Filler sentences (history, mission, slogans)

Slider from 0 sentences to 16 sentences, in steps of 2; starts at 0 sentences. Sentences with nothing the post can use. They lengthen the block and dilute it.

Try, in this order

1. Start with nothing: 4 rewrites, about 13 hours a month at 40 posts. Tick role, audience, tone and constraints: 4 sentences, 8 facts, 0 rewrites.
2. Add 16 filler sentences: 20 sentences, still 8 facts, 0.4 per sentence, and 2 rewrites come back.
3. Filler back to 0, untick tone and constraints, tick the example: 1 rewrite. An example helps; it does not replace the missing parts.

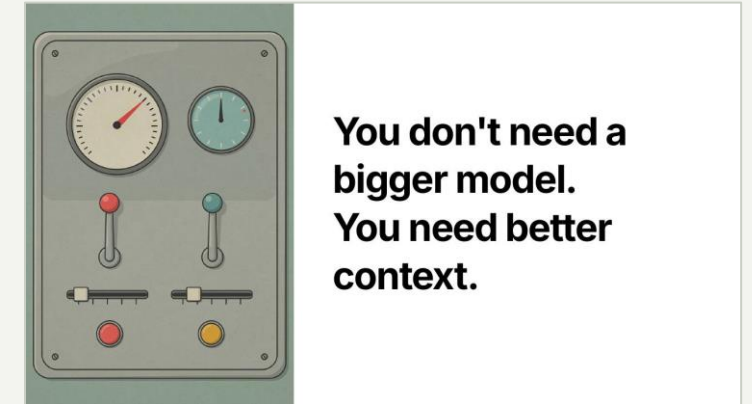
The point. Context replaces the model's defaults with your facts. What counts is how many usable facts each sentence carries, not how long the block is. Rewrite counts and minutes (5 per rewrite) are fixed teaching values chosen to show the pattern; not measurements.

Before we move on

Three things from this block

- 1 You do not need a bigger model; you need better context.
- 2 Write the block once, keep it, and reuse it for every prompt on that account.
- 3 Count facts, not sentences.

From video V21



V21, page 14 of 20. You don't need a bigger model

Context = Custom Tool



Generic Model

No context → guesses → defaults → wrong for your use case.

Custom Tool

With context → directed → targeted → professional-grade output.

V21, page 13 of 20. Context = custom tool

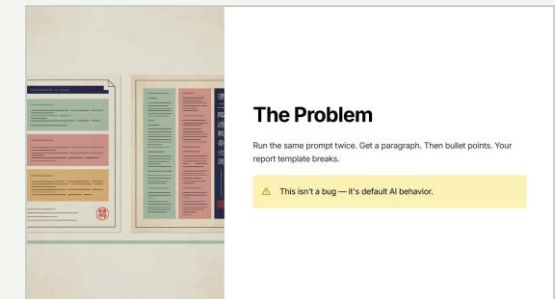
When the answer feeds a report, a sheet or another tool, specify its shape

A format block near the end of the prompt, with numbers instead of “brief”, turns a different shape every run into the same shape every time.

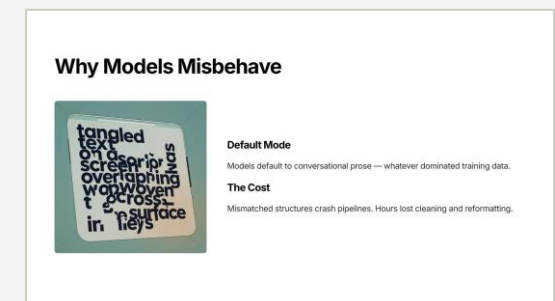
From video V22



V22, page 1 of 15. Format-forcing



V22, page 2 of 15. The problem



V22, page 3 of 15. Why models misbehave

“What are the main risks of cloud migration?” against a format block.

Loose

Three to five risks of varying length, in prose. Of the 20 answers a week that feed the risk dashboard, 6 fit its template; the other 14 are reformatted by hand.

the word: format mismatch

Forced

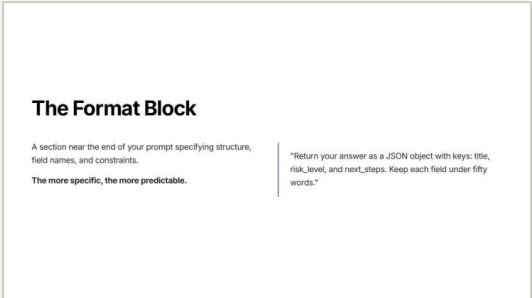
“List the top three risks as numbered items, each with a one-sentence explanation and one mitigation; each field under fifty words.” Placed at the end: 19 of 20 fit.

the word: format-forcing

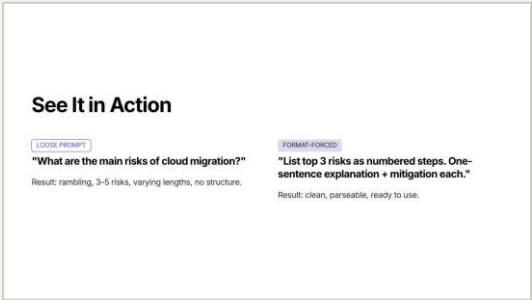
Every answer that does not fit costs someone about six minutes of reformatting. The format block costs one line in the prompt.



V22, page 5 of 15. Vague vs. forced



V22, page 7 of 15. The format block



V22, page 9 of 15. See it in action

A story: the Mars orbiter lost to a units mix-up, 1998 to 1999

1



An orbiter heads for Mars

On 11 December 1998 NASA launched Mars Climate Orbiter to study Mars from orbit. NASA's figures put the orbiter and its sister lander together at about 328 million US dollars, launches included.

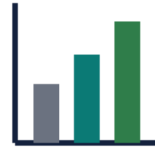
2



One file, two teams

During the flight, software used by the spacecraft team in Colorado wrote thruster data into a file for the navigation team in California. The agreed specification said metric units; the file used English units.

3



Small errors that added up

Every value came out about 4.45 times too small. The thrusters fired often, so over nine months the error grew. Navigators noticed odd results but raised them only informally.

4



Lost on arrival, 1999

On 23 September 1999 the orbiter fired its engine to enter orbit and was never heard from again: it passed an estimated 57 km above Mars, below the 80 km it could survive. A review board traced it to that one file.

5



Check the shape at the hand-off

The specification set the format and the units of that file; the file did not follow it, and no check caught the mismatch. When AI output feeds a sheet or another program, specify its shape and check it on arrival.

Sources: NASA, Mars Climate Orbiter Mishap Investigation Board Phase I Report (10 November 1999); NASA JPL, "Mars Climate Orbiter team finds likely cause of loss" (30 September 1999); NASA mission page and NSSDCA catalogue; Wikipedia, "Mars Climate Orbiter" (checked 6 October 2026); retold in our own words.

If you remember nothing else

Three things to keep

- 1 Without a format instruction a model defaults to prose, and the shape changes from run to run.
- 2 A format block near the end of the prompt: the structure, the field names, the limits in numbers.
- 3 Choose the format the next step expects: a list, a table, CSV rows, JSON, or plain text with a sentence limit.

From video V22

Key Takeaways

Specify Always Format when output feeds any system, report, or template.	Use Numbers Not "concise" — "under 50 words."
Place It Last Format block near the end of your prompt.	Start Today Rewrite one loose prompt with a format block now.

V22, page 15 of 15. Key takeaways

Common Output Formats

 Lists Bullet or numbered	 Tables Markdown or CSV rows
 JSON For APIs and tools	 Plain Text With sentence limits

Choose the format that matches what comes next. The format block removes guesswork.

V22, page 8 of 15. Common output formats

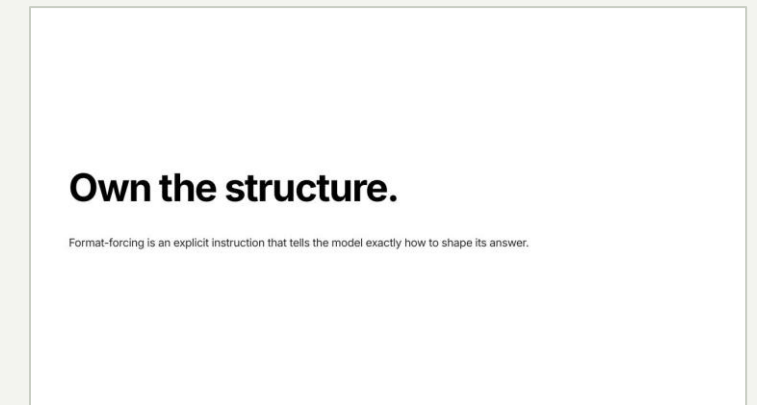
Which of these instructions is format-forced?

- A “Summarise this.”
- B “Summarise this briefly.”
- C “Summarise this in exactly three bullet points of about fifteen words each.”
- D “Summarise this well.”

From video V22



V22, page 5 of 15. Vague vs. forced



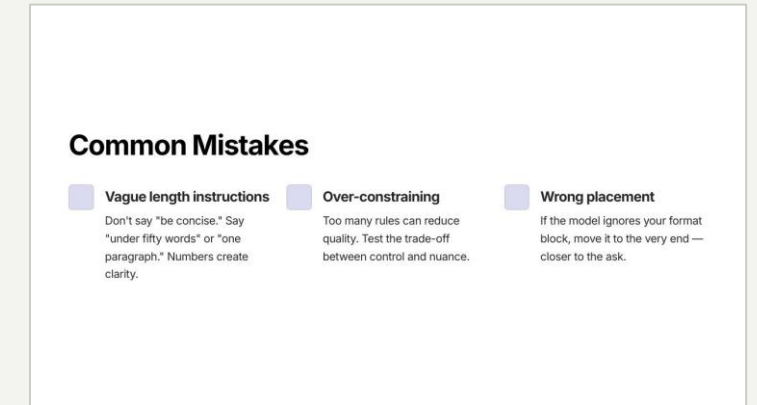
V22, page 4 of 15. Own the structure

Type the letter in the Zoom chat. The answer is discussed in the session.

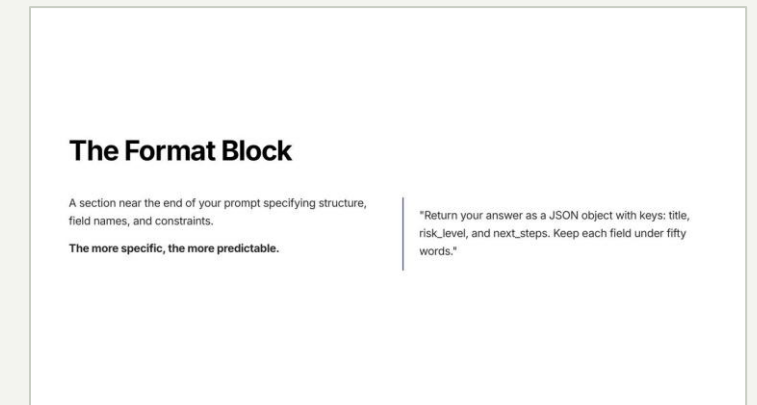
Your format block sits at the top of a long prompt, and the model keeps ignoring it. What is V22's first fix?

- A Delete the format block.
- B Move the format block to the end, next to the request, and make it more explicit; then test it.
- C Raise the temperature.
- D Switch to another model.

From video V22



V22, page 12 of 15. Common mistakes

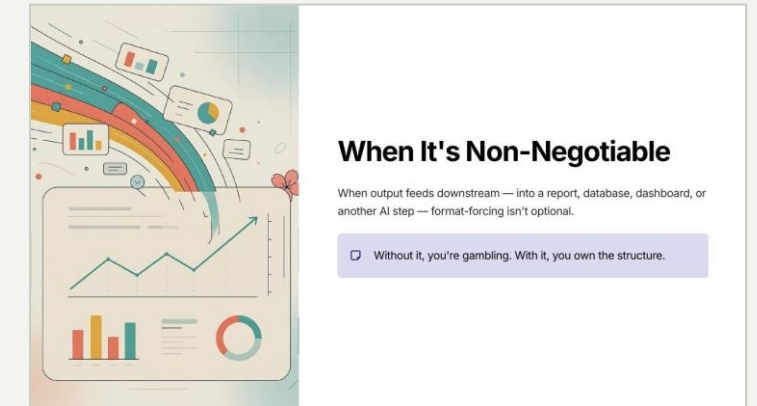


V22, page 7 of 15. The format block

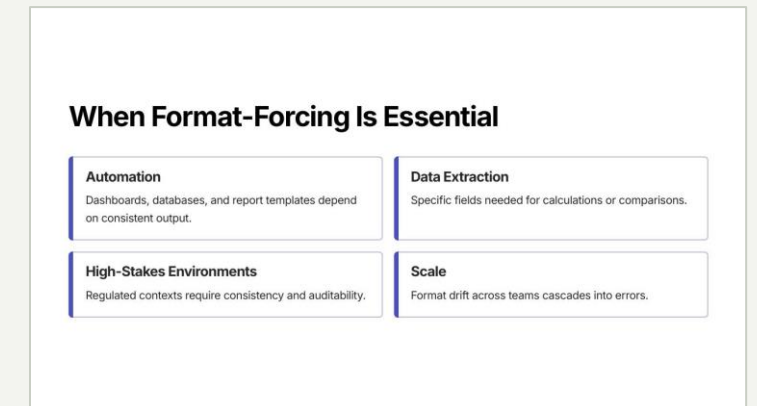
Type the letter in the Zoom chat. The answer is discussed in the session.

The output feeds a dashboard that reads three fields: title, risk_level and next_steps. Which instruction do you write?

- A “Answer in a nice structure.”
- B “Return a JSON object with the keys title, risk_level and next_steps; each field under fifty words.”
- C “Be concise.”
- D “Use bullet points.”



V22, page 6 of 15. When it's non-negotiable



V22, page 11 of 15. When format-forcing is essential

Type the letter in the Zoom chat. The answer is discussed in the session.

Your turn, described: lock the shape

The numbers are this block's dashboard: 20 AI answers a week about project risks, each fed into a template with numbered fields. Choose the format instruction, where it sits and the length rule. The panel counts the answers that fit and the minutes spent fixing the rest.

What you can change

Format instruction

Choose: No format instruction / "Be concise." / "Top three risks, numbered; one-sentence explanation and one mitigation each." / "Return JSON with the keys title, risk_level, next_steps.". The format block. The more exact it is, the less the model has to choose.

Where the format block sits

Choose: At the start of the prompt / At the end, next to the request. V22: if the model ignores the block, move it to the end. Test it: the best order varies by model.

Length rule

Choose: No length rule / "Keep it brief." / "Each field under fifty words.". Numbers create clarity; "brief" means different things to you and to the model.

Try, in this order

1. No format: 6 of 20 fit, 84 minutes of reformatting a week. Choose "Be concise.": 8 of 20; a vague word barely helps.
2. Choose the numbered list: 15 of 20 at the start; move it to the end: 18. Add "each field under fifty words": 19.
3. Choose JSON at the end with the fifty-word rule: 20 of 20, 0 minutes. Set the length rule to "brief": back to 19.

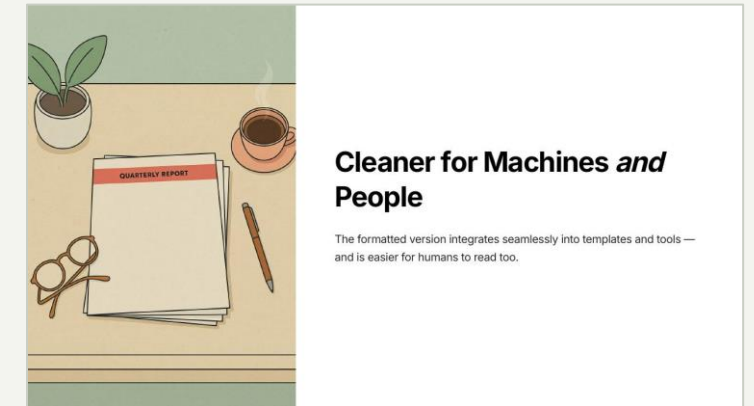
The point. When output feeds something else, its shape is part of the job. Say exactly what shape, with numbers, and put it where the model reads it last. Fit counts are fixed teaching values chosen to behave like typical runs; models still miss exact counts sometimes, so check what arrives.

Before we move on

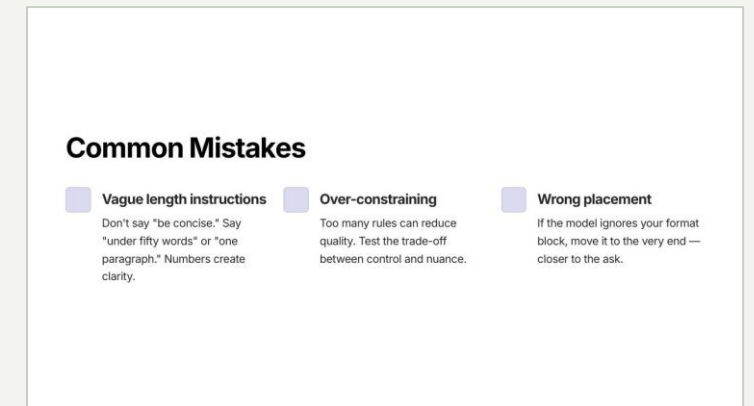
Three things from this block

- 1 Format-forcing helps people too: a structured answer is faster to read.
- 2 Too many rules can cost nuance; test the trade-off.
- 3 Rewrite one loose prompt with a format block this week, and log both versions.

From video V22



V22, page 10 of 15. Cleaner for machines and people

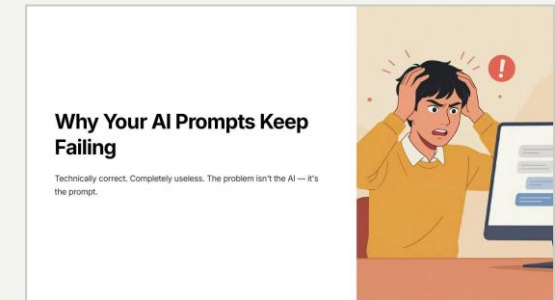


V22, page 12 of 15. Common mistakes

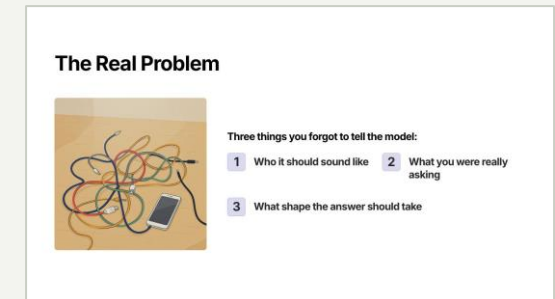
Three named patterns fix three failures: persona, question refinement, format-forcing

Wrong tone, a fuzzy question, reformatting every time: name the frustration, then apply the matching pattern.

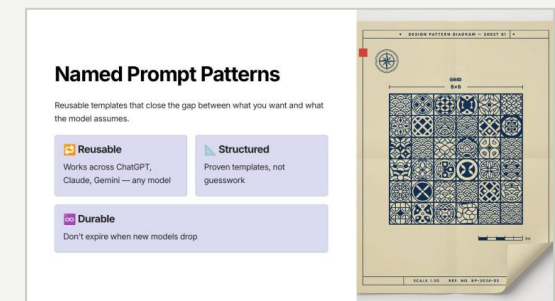
From video V23



V23, page 1 of 20. Why your AI prompts keep failing



V23, page 2 of 20. The real problem



V23, page 3 of 20. Named prompt patterns

“How do I improve my model?” has four meanings. The model picks one.

Without a pattern

It could mean fixing errors, better input data, tuning settings or a new design. The model answers one of them, fluently. In our example, about 20 minutes go on the wrong problem.

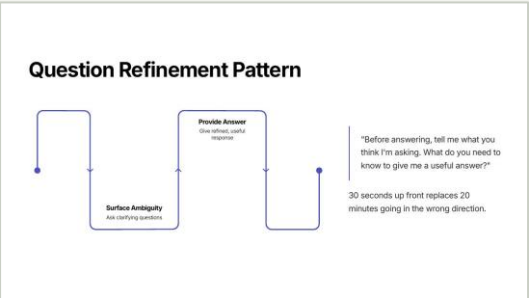
the word: ambiguity

With question refinement

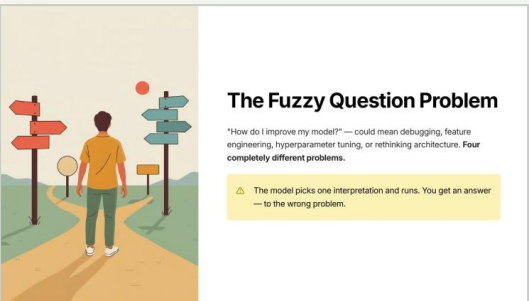
“Before answering, tell me what you think I am asking, and what you need to know to answer it well.” About 30 seconds up front, and the right problem gets answered.

the word: question refinement

Persona fixes the voice, question refinement fixes the problem being solved, format-forcing fixes the shape. Used together in one prompt, the effects add up.



V23, page 9 of 20. Question refinement pattern



V23, page 8 of 20. The fuzzy question problem

Persona: Before & After

✗ Weak Prompt "Write an explanation of machine learning." Model guesses tone, audience, depth — usually lands on generic.	✓ With Persona "You are a data scientist explaining machine learning to business stakeholders with no technical background. Under 200 words."
--	--

Three things changed: **role, audience, constraint**. The model shifts register, vocabulary, and assumed knowledge — all at once.

V23, page 5 of 20. Persona: before and after

A story: how good solutions got names, 1977 to 2023

1



Patterns for towns and rooms

In 1977 the architect Christopher Alexander and colleagues published “A Pattern Language”: 253 named solutions to problems that keep coming up in towns and buildings. Number 159 is “Light on two sides of every room”.

2



Programmers borrow the idea

In 1987 Kent Beck and Ward Cunningham applied the idea to software, writing five patterns for screens, such as “Window per task”. People new to the system designed sensible screens with them after a day of practice.

3



The Gang of Four, 1994

In 1994 Erich Gamma, Richard Helm, Ralph Johnson and John Vlissides published “Design Patterns”, a catalogue of 23 named solutions to common software design problems. The four became known as the Gang of Four.

4



Patterns for prompts, 2023

In February 2023 Jules White and colleagues at Vanderbilt University published a catalogue of prompt patterns. It named 16, including “Persona” and “Question Refinement”, and showed how to combine them.

5



A name others can reuse

Each time, a solution that kept coming up got a name and a short description. That lets a colleague reuse it, combine it, and understand you at once when you say “persona plus question refinement”.

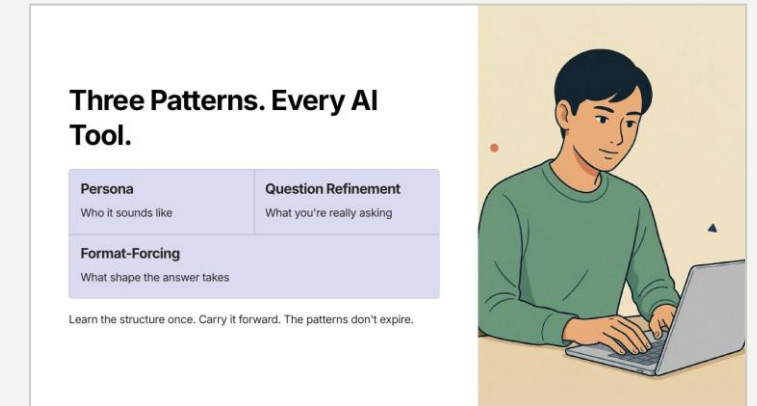
Sources: C. Alexander et al., A Pattern Language (Oxford University Press, 1977) and patternlanguage.com; K. Beck and W. Cunningham, “Using pattern languages for object-oriented programs” (1987); InformIT, Design Patterns (1994); J. White et al., “A prompt pattern catalog to enhance prompt engineering with ChatGPT”, arXiv (2023) (checked 6 October 2026); retold in our own words.

If you remember nothing else

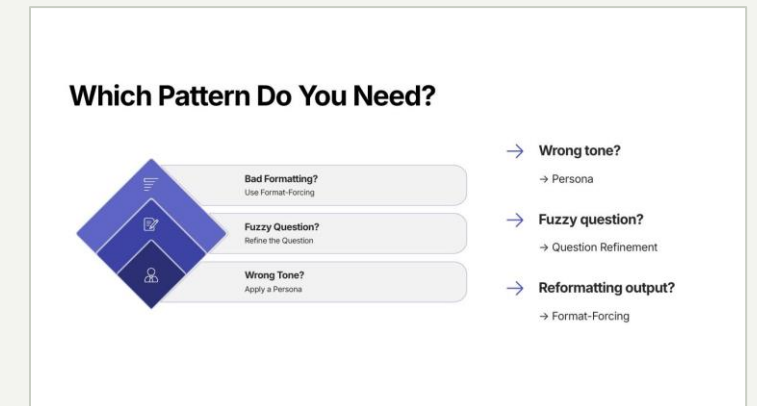
Three things to keep

- 1 Persona: tell the model who it is and who it is writing for; it fixes tone and register.
- 2 Question refinement: ask it to restate your question and say what it needs first; it fixes the fuzzy question.
- 3 Format-forcing: lock the structure; it fixes reformatting. Combine them and the effects add up.

From video V23



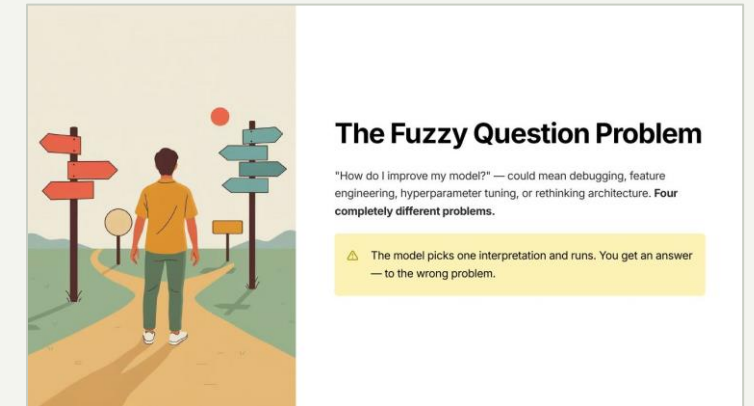
V23, page 20 of 20. Three patterns, every AI tool



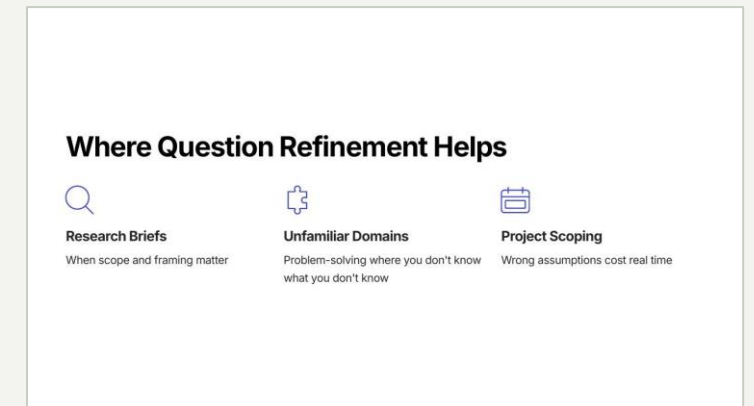
V23, page 16 of 20. Which pattern do you need?

A marketing student asks: “How do I improve our campaign’s results?” It could mean budget, targeting, creative or timing. Which pattern fits?

- A Persona.
- B Question refinement.
- C Format-forcing.
- D None: ask a shorter question.



V23, page 8 of 20. The fuzzy question problem



V23, page 10 of 20. Where question refinement helps

Type the letter in the Zoom chat. The answer is discussed in the session.

Support replies for a clothing retailer come back stiff and full of technical words. Which pattern, and what would you write?

- A Persona: “You are a friendly support agent for a clothing brand, writing to customers with no technical background.”
- B Question refinement: “Before answering, tell me what you think I am asking.”
- C Format-forcing: “Use a table.”
- D Temperature: set it higher.

Persona: Before & After

✗ Weak Prompt

“Write an explanation of machine learning.”

Model guesses tone, audience, depth — usually lands on **generic**.

✓ With Persona

“You are a data scientist explaining machine learning to business stakeholders with no technical background. Under 200 words.”

Three things changed: **role, audience, constraint**. The model shifts register, vocabulary, and assumed knowledge — all at once.

V23, page 5 of 20. Persona: before and after

Where Persona Shows Up



Customer Service

Replies that match brand voice



Documentation

Pitched to the right reader level



Sales Pitches

Speaks the buyer's language, not the product team's

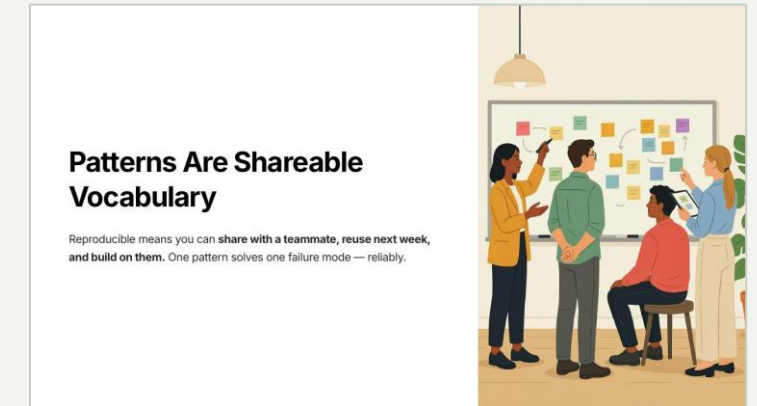
V23, page 6 of 20. Where persona shows up

Type the letter in the Zoom chat. The answer is discussed in the session.

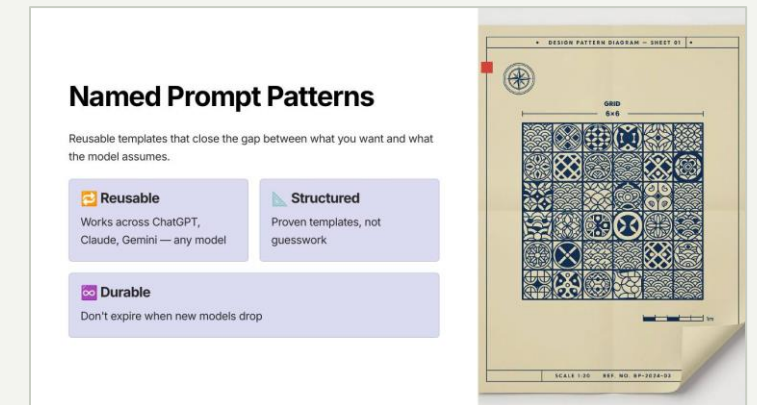
Why give prompt patterns names at all?

- A Names sound professional.
- B A shared vocabulary: a teammate can reuse, combine and build on a pattern, and it works in any tool, whatever model comes next.
- C The model only responds to named patterns.
- D To patent them.

From video V23



V23, page 17 of 20. Patterns are shareable vocabulary



V23, page 3 of 20. Named prompt patterns

Type the letter in the Zoom chat. The answer is discussed in the session.

Your turn, described: match the pattern to the frustration

The numbers are this block's working week: prompts that suffer from all three failures. Choose how many prompts you write a week and switch each pattern on. The panel shows what each failure costs and what each pattern saves.

What you can change

Prompts you write a week

Slider from 5 prompts to 50 prompts, in steps of 5; starts at 10 prompts. Each one suffers from the three failures until a pattern fixes it.

Persona: "You are a data scientist explaining to managers..."

On or off; starts off. Fixes the wrong tone. Saves about 10 minutes of rewriting a prompt.

Question refinement: "First tell me what you think I am asking."

On or off; starts off. Fixes the fuzzy question. Costs 30 seconds; saves about 20 minutes on the wrong problem.

Format-forcing: "A table: risk, severity 1 to 5, mitigation."

On or off; starts off. Fixes the shape. Saves about 8 minutes of reformatting a prompt.

Try, in this order

1. 10 prompts a week, no pattern: 38 minutes lost a prompt, 380 a week. Tick question refinement: 18.5 a prompt, 185 a week: 30 seconds up front saves 19.5 minutes.
2. Tick persona and format-forcing too: 0.5 minutes a prompt, 5 a week. The three patterns combine.
3. Untick all three and move to 50 prompts a week: 1,900 minutes, about 127 hours a month.

The point. A named pattern solves one kind of failure, every time, in any tool, and a teammate knows what you mean when you say its name.

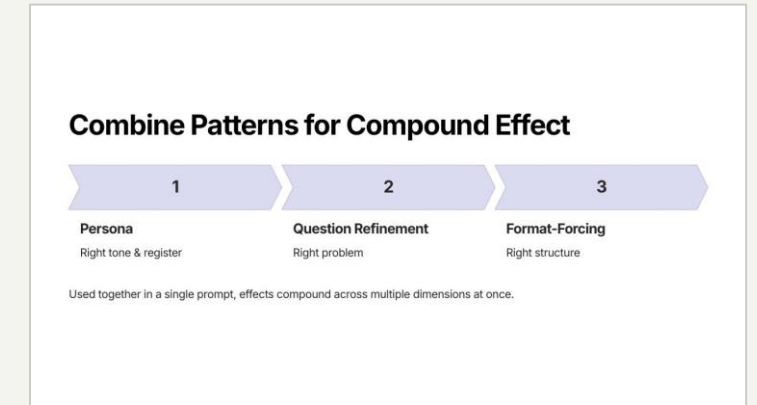
Minutes per failure are fixed teaching values (V23 itself quotes 30 seconds against 20 minutes as an illustration); not measurements.

Before we move on

Three things from this block

- 1 Pick the pattern that matches your frustration, not the one you like best.
- 2 Practise on one real prompt: original, pattern, revised; run both and compare tone, structure and relevance.
- 3 Patterns outlive models: learn the structure once.

From video V23



V23, page 18 of 20. Combine patterns for compound effect

Your Practice Task

Pick a real prompt you use repeatedly — drafting emails, summarizing docs, analyzing data.

01 _____

Write the original prompt

02 _____

Choose the pattern that matches your frustration

03 _____

Write the revised prompt

04 _____

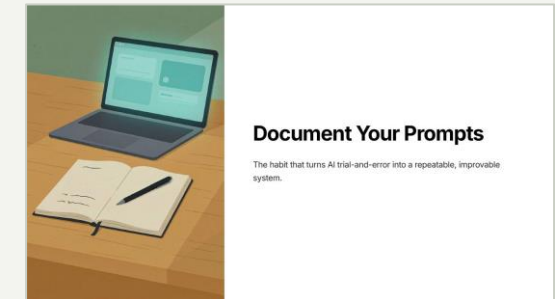
Run both — compare tone, structure, relevance

V23, page 15 of 20. Your practice task

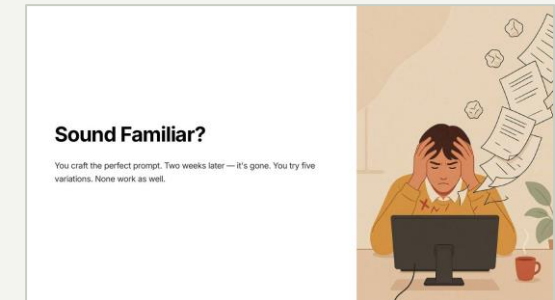
A prompt that worked is only as useful as the record you kept of it

The exact prompt, the model and settings, the date, the output, a version number and a one-line change note: about two minutes a prompt.

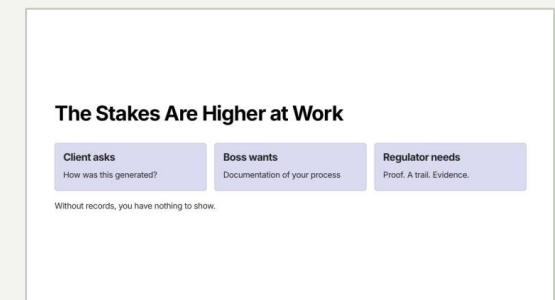
From video V24



V24, page 1 of 25. Document your prompts



V24, page 2 of 25. Sound familiar?



V24, page 3 of 25. The stakes are higher at work

Two weeks later, the client asks: how was this generated?

No record

You remember roughly what you typed.
Five tries to rebuild it, none as good:
about 45 minutes lost, and nothing to
show the client.

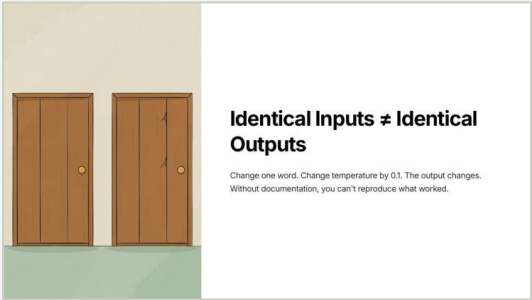
the word: reproducibility

A log row

“v2 · 5 June · GPT-4 · temperature 0.5 ·
clear, concise · rewrote for tone.” Rerun in
two minutes, and a dated trail to show.

the word: audit trail

Documenting takes about two minutes a prompt. Rebuilding from memory takes far longer, and still gives a different answer.



V24, page 4 of 25. Identical inputs ≠ identical outputs

Organize in a Table

Put versions side by side. Spot which tweaks improve quality instantly.

Version	Date	Model	Temp	Output Summary	What Changed
v1	Jun 1	GPT-4	0.7	Formal, lengthy	Initial draft
v1.1	Jun 3	GPT-4	0.7	Shorter, lost context	Added word limit
v2	Jun 5	GPT-4	0.5	Clear, concise ✓	Rewrote for tone

V24, page 14 of 25. Organise in a table

The Cost of Skipping It

2 minutes To document a prompt properly	Hours Lost rebuilding from memory later
---	---

V24, page 21 of 25. The cost of skipping it

A story: the penicillin paper that Oxford found, 1928 to 1945

1



A mould in a dish, 1928

In September 1928 Alexander Fleming came back from holiday to his laboratory at St Mary's Hospital in London. In one dish of bacteria a mould had grown, and the bacteria around it had died.

2



Written up in 1929

He named the substance penicillin and described his work in the British Journal of Experimental Pathology in June 1929. He could not purify it, and he set that work aside.

3



Found again in Oxford

In 1938 Ernst Chain and Howard Florey at Oxford began studying substances that kill bacteria. Chain found Fleming's paper, and the lab still had a sample of his mould. In May 1940 infected mice given penicillin lived.

4



A first patient, then a prize

On 12 February 1941 Albert Alexander, an Oxford police constable, was the first person treated with penicillin for an infection. He improved, but the supply ran out. In 1945 the three men shared the Nobel Prize.

5



A record someone could use

Oxford started from two things Fleming left: his paper and a saved sample. Because he had recorded what he did and saw, others could check it and go further. A precise prompt log lets a colleague do the same.

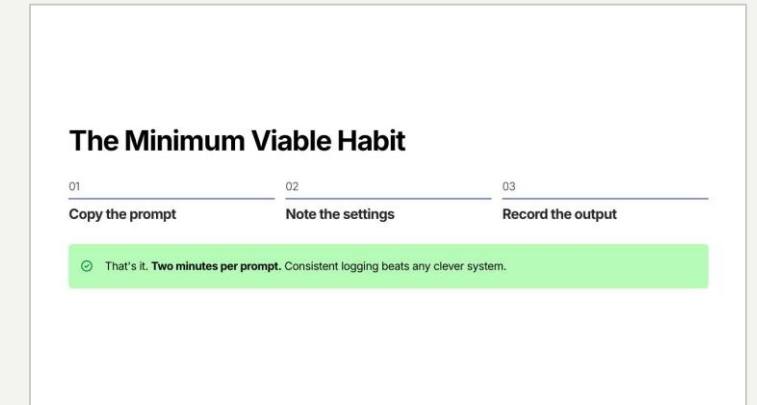
Sources: The Nobel Prize, "The Nobel Prize in Physiology or Medicine 1945"; Science History Institute, "Alexander Fleming" and "Howard Walter Florey and Ernst Boris Chain"; University of Oxford, "Penicillin: the Oxford story"; Wikipedia, "Discovery of penicillin" (checked 6 October 2026); retold in our own words.

If you remember nothing else

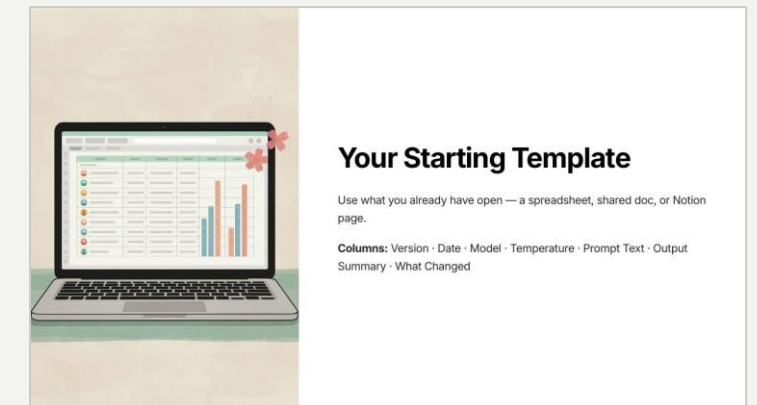
Three things to keep

- 1 Record the exact prompt, the model and its settings, the date and time, the output or a one-line summary, and a version number with a change note.
- 2 Number versions (v1, v1.1, v2) and keep them side by side in a table, with a notes column.
- 3 Log every prompt, not only the ones that feel important; review the log for five minutes a week.

From video V24



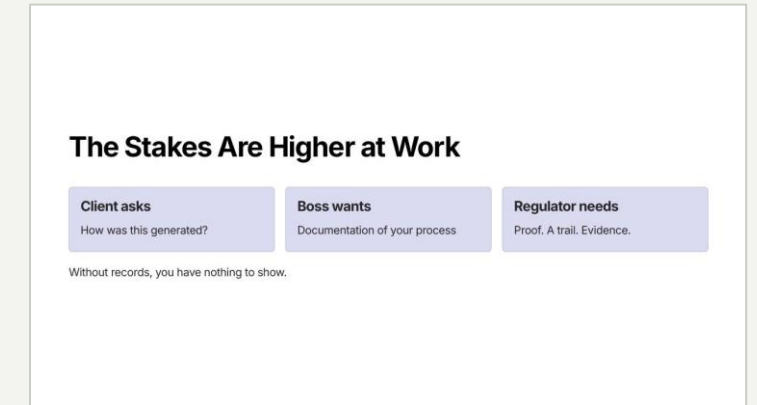
V24, page 24 of 25. The minimum viable habit



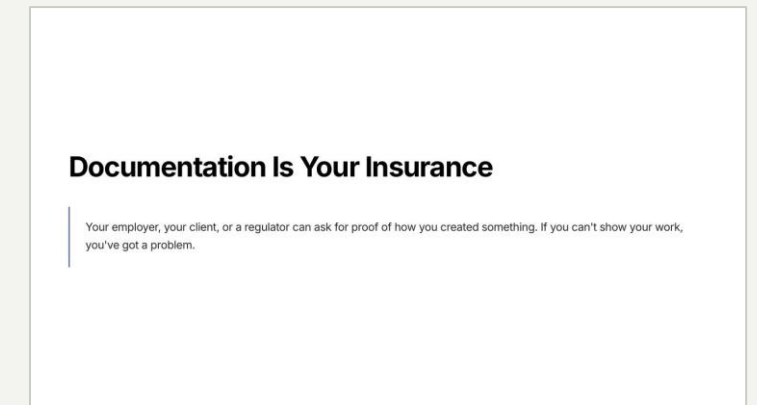
V24, page 16 of 25. Your starting template

A client asks: “How was this report summary generated?” Which answer protects you?

- A “I used ChatGPT.”
- B The log row: the exact prompt, the model and its version, the settings, the date, an output summary, the version number.
- C A screenshot of the final text.
- D “That is confidential.”



V24, page 3 of 25. The stakes are higher at work



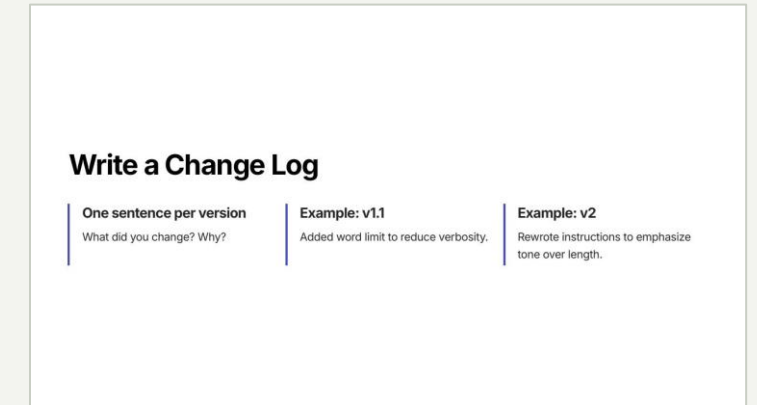
V24, page 5 of 25. Documentation is your insurance

Type the letter in the Zoom chat. The answer is discussed in the session.

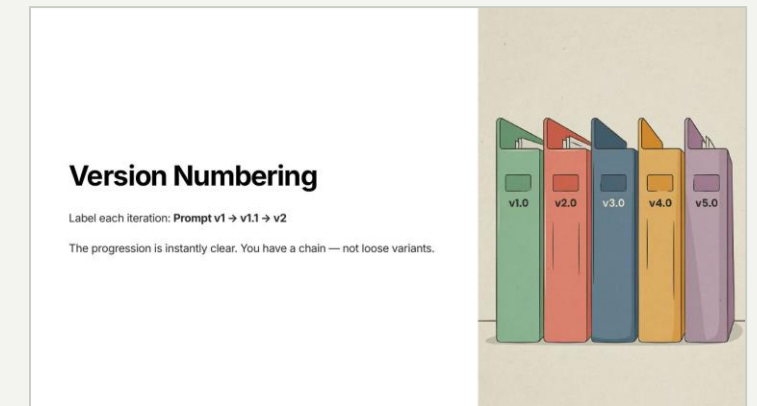
In your log, the entry for version v1.1 reads: “Added a word limit to reduce wordiness.” Which part of the log is that?

- A The output.
- B The change log: one sentence on what changed and why.
- C The model setting.
- D The timestamp.

From video V24



V24, page 13 of 25. Write a change log



V24, page 12 of 25. Version numbering

Type the letter in the Zoom chat. The answer is discussed in the session.

In the deck’s own table, version v2 rewrote the wording and moved the temperature from 0.7 to 0.5. Its note says only “Rewrote for tone.” What does V20’s method say about this entry?

- A Nothing wrong: the output improved.
- B Two variables changed at once, so the log cannot say which one made v2 better; the note should name both, and a clean test changes one.
- C The temperature does not matter.
- D The model changed too.

Type the letter in the Zoom chat. The answer is discussed in the session.

Organize in a Table

Put versions side by side. Spot which tweaks improve quality instantly.

Version	Date	Model	Temp	Output Summary	What Changed
v1	Jun 1	GPT-4	0.7	Formal, lengthy	Initial draft
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v2	Jun 5	GPT-4	0.5	Clear, concise ✓	Rewrote for tone

V24, page 14 of 25. Organise in a table



Identical Inputs ≠ Identical Outputs

Change one word. Change temperature by 0.1. The output changes.
Without documentation, you can't reproduce what worked.

V24, page 4 of 25. Identical inputs ≠ identical outputs

Your turn, described: could you reproduce it in two weeks?

The numbers are this block's log: the customer-service prompt from the deck (v1, v1.1, v2) and your weekly volume. Tick the fields you record. The panel says whether v2 can be rerun and explained, and compares minutes documenting with minutes rebuilding.

What you can change

The exact prompt text, copied and pasted

On or off; starts off. Every word and line break. A paraphrase loses the phrasing that made it work.

The model name and version

On or off; starts off. The same prompt on a different model gives a different result.

The settings (temperature and any others)

On or off; starts off. A change of 0.1 in temperature changes the output; write "default" if your tool has no setting.

The date and time of the run

On or off; starts off. Places the run in your work; models are updated over time.

The output, or a one-line summary of it

On or off; starts off. Without the output you cannot tell later whether the prompt worked.

A version number and a change note (v1.1: added a word limit)

On or off; starts off. Turns loose variants into a chain you can compare.

Prompts you write a week

Slider from 5 prompts to 40 prompts, in steps of 5; starts at 20 prompts. About 1 in 5 is needed again later; each one rebuilt from memory costs about 45 minutes.

Try, in this order

1. Nothing recorded, 20 prompts a week: 0 minutes documenting, 180 rebuilding (4 prompts, 45 minutes each). v2 cannot be rerun.
2. Tick all six: 40 minutes documenting, 8 rerunning, 48 in total; v2 reruns, and the client gets a dated trail.
3. Untick the settings: the same words on unknown settings give a different output, and rebuilding climbs to 80 minutes.

The point. The log is your laboratory notebook: it lets you rerun, explain and improve. Two minutes now saves the time spent trying to remember. Minutes are fixed teaching values (the deck's own "two minutes against hours" is an illustration); not measurements.

Before we move on

Three things from this block

- 1 Two minutes a prompt is the price of being able to rerun, explain and improve.
- 2 Your Assignment 1 log is this table: four variants, one element each, every run recorded.
- 3 Keep one log file for the whole course; every session from now on adds to it.

From video V24

From Luck to Skill



Version logs turn random experimentation into something **systematic**.
Track what changed. See what happened. Patterns emerge. Skill replaces luck.

V24, page 22 of 25. From luck to skill

Start Now

Log every prompt from this point forward. The habit of consistent documentation is more valuable than any system you could design.

2 MIN/PROMPT REPRODUCIBLE RESULTS SYSTEMATIC IMPROVEMENT



V24, page 25 of 25. Start now

One prompt, one change at a time, and the differences in the chat

The baseline prompt (copy it exactly)

You are a marketing copywriter. Write a launch announcement for StudyCents, a free budgeting app for university students in Canada. It links to a bank account, sends a weekly spending summary every Sunday and launches on 1 November 2026. Write one paragraph of at most 80 words.

Three runs. **Run 0**, zero-shot: “Write an announcement for a budgeting app.” **Run 1**, the baseline above: role, instruction, context, format. **Runs 2 and 3**: the baseline with exactly one element changed each time.

StudyCents is a fictional app invented for this exercise; any real app with that name is unrelated.

Change the role, then the format; never both at once

- 1 I run Run 0 and Run 1 on my screen and paste both into the chat. You run them in your own tool, in a fresh conversation each time.
- 2 Run 2, role only: replace “You are a marketing copywriter” with “You are a financial analyst”. Every other word stays the same. Type in the chat one thing that moved.
- 3 Run 3, format only: back to the copywriter; replace the last sentence with “Present it as a three-column table: feature, what it does, why it matters to a student.” Type one thing that moved.
- 4 Last, in one line: which element would you change next, and why? This is the model for Assignment 1, the Four-Variation Prompt Log, due Monday 12 October at 23:59.

Log every run as you go: the exact prompt, the tool and model, the date, the output (V24). If your tool shows no model name or temperature, write “default”.

Sixteen hours with the wrong instructions

On 4 July 2025 Elon Musk said Grok, xAI's chatbot on X, had been improved "significantly". On 6 July xAI added a line to Grok's public instructions: answers "should not shy away from making claims which are politically incorrect, as long as they are well substantiated." On 8 July the bot posted a wave of antisemitic replies, praising Hitler and calling itself "MechaHitler". That evening xAI froze the account, began deleting posts and removed the line.

On 12 July xAI apologised. Its account: a code change upstream of the bot, active for 16 hours, had switched old instructions back on, among them "You tell it like it is and you are not afraid to offend people who are politically correct" and "Reflect that in your response" about the tone of each post. The bot then mirrored extremist posts in the threads it answered. xAI said the cause was "independent of the underlying language model"; in August its legal head told lawmakers that no model parameters, training data or fine-tuning had changed.

Poll, type A or B in the chat. A: a prompt problem: the instructions changed and the output followed. B: a data problem: the instructions only released what the training data and the posts already contained.

Four questions by name, assigned sides, a re-vote

Since then. July was the third time in 2025 that xAI blamed an unreviewed or unintended change to Grok's instructions. From late December 2025 its image tools were used to make non-consensual sexualised images of real people, including minors, according to regulators. Ofcom, the European Commission, the UK and Irish data regulators and California's Attorney General opened cases; in June 2026 Canada's Privacy Commissioner found that X and xAI broke Canadian privacy law. The other cases were still open at their last updates.

- 1 xAI says the model and its training data never changed; only instructions did. Using V18 and V20: what does that say about how far one prompt element can move an output?
- 2 Using V13 and V14: the bot mirrored the posts in the threads it read. Where did the patterns it repeated come from, if not from the instructions?
- 3 Three times in one year the explanation was a change nobody reviewed. Using V17 and V24: which two habits from today would have caught it, and at which step?
- 4 You run customer service at a firm that uses Grok in its chat window. It is 9 July 2025. What do you do first: pause the bot, check your own instructions, or wait for xAI's explanation?

Debate: odd numbers in the participant list argue A, even numbers B; two minutes to type one argument, two speakers per side, then the same poll again. Sources (checked 6 October 2026): xAI's statement of 12 July 2025 (CNN, NBC News); xAI's prompt repository on GitHub; the House letter of 10–11 July and xAI's August reply (Jewish Insider); Privacy Commissioner of Canada, 11 June 2026; Ofcom, European Commission and ICO case pages.

What to do before Session 3, and what is assessed

Before Friday 9 October 2026

- 1 Bias Audit (Module 3): profiles of a “typical” financial analyst, social media manager and entrepreneur; annotate assumptions and word choices; 200-word reflection on a risk in your sector. Before Friday 9 October.
- 2 Go / No-Go Decision Worksheet (Module 5), in pairs: eight cards, go or no-go with one sentence each, disagreements settled in writing. Before Friday 9 October.
- 3 Quizzes 3 and 4 on the platform before Friday 9 October.
- 4 Assignment 1, Four-Variation Prompt Log (Module 4): one task, four variants each changing one element, outputs in a table, a paragraph on the biggest shift. Due Monday 12 October, 23:59.
- 5 Watch V25 to V35 (about 66 minutes) and do the Module 5 and 6 readings; Session 3 assumes you have.

How the grade is built, with the dates

Participation	20%	every session, full / half / none
Quizzes and portfolio	20%	Quizzes 3–4 by 9 Oct, 5 by 14 Oct; A1 12 Oct, A2 13 Oct, A3 16 Oct, A4 20 Oct, A5 23 Oct
Peer learning	10%	workflow-map review in Session 5; capstone forms by 30 Oct
Capstone portfolio	50%	draft and recording 28 Oct; final Wednesday 4 November 2026

Next: Session 3: Friday 9 October 2026, 10:30–12:30 EDT. When not to use AI, and catching hallucinations before they catch you (V25 to V35).